

Review

Advances in UAV Path Planning: A Comprehensive Review of Methods, Challenges, and Future Directions

Wenlong Meng ^{1,*} , Xuegang Zhang ², Lvzhuoyu Zhou ¹, Hangyu Guo ¹ and Xin Hu ¹¹ School of Computer Science and Technology, Harbin Institute of Technology, Weihai 264209, China² School of Mechanical, Electrical & Information Engineering, Shandong University, Weihai 264209, China

* Correspondence: mengwenlong_hit@163.com

Abstract: Unmanned aerial vehicles (UAVs) have revolutionized fields such as monitoring, cargo delivery, precision farming, and emergency response, demonstrating remarkable flexibility and operational effectiveness. A fundamental aspect of UAV autonomy lies in route optimization, which determines efficient paths while considering factors like mission goals, safety, and power consumption. This article presents an extensive overview of methodologies for UAV route planning, including deterministic models, stochastic sampling techniques, biologically inspired methods, and integrated algorithmic frameworks. The discussion extends to their performance in various operational contexts, including stationary, moving, and three-dimensional settings. Innovative methods utilizing artificial intelligence, particularly machine learning and neural networks, are emphasized for their promise in facilitating adaptive responses to intricate, evolving environments. Furthermore, strategies focused on reducing energy usage and enabling coordinated operations among multiple drones are analyzed, addressing issues such as prolonged operation, distribution of assignments, and navigation around obstacles. Although notable advancements have been achieved, challenges like high computational demands and the need for immediate responsiveness persist. By consolidating the latest progress, this survey provides meaningful perspectives and guidance for the ongoing evolution of UAV route planning solutions.



Academic Editors: Heng Shi, Jihong Zhu, Zheng Chen and Minchi Kuang

Received: 25 March 2025

Revised: 1 May 2025

Accepted: 8 May 2025

Published: 16 May 2025

Citation: Meng, W.; Zhang, X.; Zhou, L.; Guo, H.; Hu, X. Advances in UAV Path Planning: A Comprehensive Review of Methods, Challenges, and Future Directions. *Drones* **2025**, *9*, 376. <https://doi.org/10.3390/drones9050376>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: unmanned aerial vehicle; path planning; motion control; remote sensing applications; cooperative navigation

1. Introduction

1.1. Background

1.1.1. Advancements in UAV Technology Development

In today's age of swift technological advancement, unmanned aerial vehicles (UAVs) have become crucial tools within contemporary engineering and technology, acting as powerful catalysts for innovation in numerous fields. Their broad utilization not only demonstrates their exceptional adaptability and operational effectiveness but also signals their capacity to fundamentally reshape sectors such as logistics, surveillance, environmental assessment, and precision farming. Thanks to features like outstanding maneuverability, intuitive handling, and economic benefits, UAVs have established themselves as indispensable resources across multiple industries. Differing from traditional piloted aircraft, UAVs reduce operator risk by removing the necessity for onboard human control, thereby raising the level of operational safety. Additionally, notable developments in core areas—including navigation systems [1], communication frameworks [2,3], environmental data acquisition [4,5], and energy management [6]—have significantly broadened UAVs' operational

scope, enabling them to tackle a wider variety of intricate tasks with enhanced accuracy and productivity. As a result, UAVs are now widely employed in an array of industries, as depicted in Figure 1, such as city development [7], smart agriculture [8,9], emergency response [10,11], defense applications [12,13], and supply chain management [14]. Their adaptability in these domains not only shows their transformative role but also emphasizes their increasing significance in overcoming multifaceted problems.

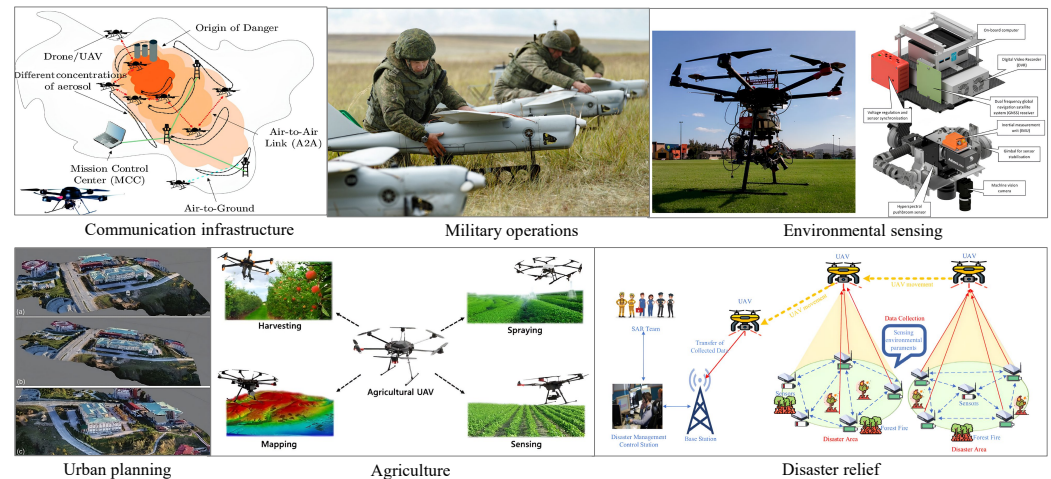


Figure 1. Extensive applications of UAVs across a broad spectrum of sectors.

1.1.2. The Significance and Challenges of UAV Path Planning

Unmanned aerial vehicles are predominantly deployed to evaluate risks and undertake dangerous operations in obstacle-rich settings, thereby removing direct threats to human personnel. Achieving effective and independent UAV performance heavily depends on accurate route planning. This process entails charting the most advantageous path from the point of origin to the target location, guaranteeing both operational safety and mission efficiency [15–17]. Figure 2 illustrates the entire UAV route planning procedure, outlining its main phases and essential elements for a comprehensive understanding of the underlying workflow. When planning UAV routes, the central aim is to create a flight path that fulfills specific mission requirements, respects environmental limitations, and seeks to optimize energy consumption, among other vital parameters [18–21]. Well-designed trajectory planning not only reduces travel duration and power expenditure but also boosts the overall quality of mission accomplishment. In addition, it is crucial for maintaining UAV safety and dependability during operations, as it lowers the risk of malfunctions and increases the probability of successful task completion.

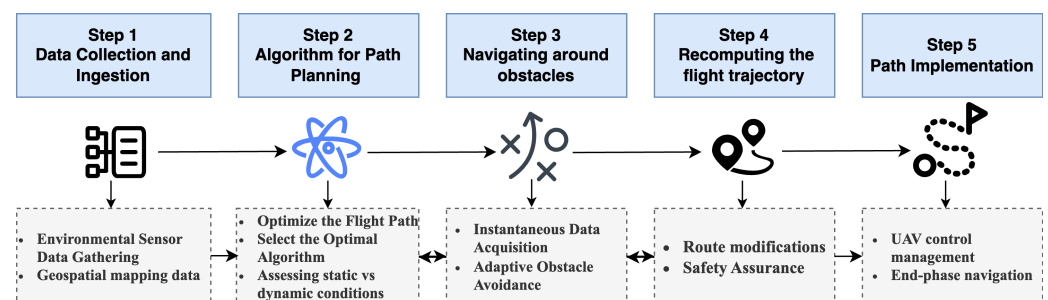


Figure 2. An overview of UAV path planning with critical stages and core elements.

Within the realm of route planning, UAVs are often required to function in settings filled with diverse obstacles, and the intricacy of these surroundings plays a crucial role in shaping the planning strategy. Such operational environments are generally divided into

two main categories: static and dynamic [22–24]. Static scenarios are marked by stationary, predictable barriers, whereas dynamic scenarios feature moving objects, necessitating on-the-fly modifications to the UAV's course for safe and effective maneuvering.

A static environment is characterized by obstacles whose locations and dimensions are either already known or can be reliably predicted beforehand. In these cases, UAVs navigate around permanent structures—such as high-rise buildings in urban areas or rugged mountain landscapes [25,26]—that do not change position over time. In contrast, dynamic environments are distinguished by the presence of obstacles that move or whose behaviors are unpredictable, making the context inherently uncertain. Examples include mobile entities like vehicles [27] and unpredictable objects such as birds in flight [28]. Operating in such dynamic settings demands that UAVs be equipped with real-time detection systems and swift decision-making mechanisms, enabling them to continually adjust their routes and maintain both safety and operational efficiency amid evolving circumstances.

Beyond external environmental obstacles, UAVs face internal challenges as well, most notably restrictions related to energy consumption. Striking an effective compromise between mission length and power efficiency remains a key focus within current research. Because UAVs predominantly rely on batteries, their operational range is fundamentally constrained by both payload limitations and advances in energy storage systems [29]. Investigations into UAV energy usage commonly address both the hardware and software domains. Regarding hardware, advancements in battery design seek to increase flight duration and decrease overall mass. On the software front, refining flight path algorithms and enhancing communication protocols can significantly reduce energy expenditure, allowing UAVs to accomplish tasks more effectively while maintaining safety and performance standards. For instance, the study in [30] presents an all-encompassing model for energy consumption derived from real-world battery data, aiming to maximize energy efficiency during UAV deployments. Moreover, additional research [22] highlights the importance of improving energy effectiveness in UAV communications, summarizing various up-to-date methods for minimizing power usage in this context. Collectively, these initiatives underscore the continuous progress in boosting UAV energy efficiency, tackling both hardware and communication-related limitations to broaden operational potential.

Collaborative operations involving multiple UAVs have become a central theme in UAV research, reflecting the growing shift from single-drone missions to coordinated multi-UAV deployments. Compared to using just one UAV, employing several drones together brings notable benefits, such as improved operational speed—especially valuable in fields like agriculture [31]—as well as the capacity to execute parallel tasks, offer redundancy, and exploit diverse strengths within the fleet [32]. Nevertheless, beyond the inherent limitations faced by individual UAVs, coordinating multiple drones introduces further challenges. These include assigning tasks efficiently [33–35] and managing interactions between flight paths [36,37] among the UAVs involved. Successfully addressing these issues is essential for maximizing mission performance and achieving seamless collaboration among UAVs, particularly when operating in complex and changing environments.

1.2. Motivation

The objective of this paper is to provide a comprehensive review of the current state of research in UAV path planning, tracing its evolution from early classical graph-theoretic algorithms to the recent advancements in cutting-edge methods. In this way, we present a thorough analysis of the key research findings and emerging trends across various techniques. UAV path planning must address a diverse range of environmental types, including, but not limited to, static environments, dynamic environments, complex terrains, urban airspace, and unknown conditions [38–40]. In this context, this paper categorizes and

critically examines existing path planning algorithms according to the specific characteristics of these environments. Each algorithm is evaluated across key dimensions such as adaptability, robustness, and limitations [41], offering a scientific foundation for selecting the most suitable path planning methods tailored to different application scenarios.

Despite significant advancements in UAV path planning, several technical bottlenecks remain in practical applications. For instance, the computational complexity of algorithms continues to be a major challenge in large-scale or high-dimensional environments [42], and existing path planning methods often show limited adaptability to highly uncertain environments in real-time application scenarios [43]. Consequently, this paper aims to explore the future directions of path planning algorithms, with the objective of offering a feasible research framework and practical guidance for technological breakthroughs and the future prospects of UAV path planning applications.

The paper is structured as follows: Section 2 introduces the UAV path planning problem, encompassing key classifications, constraints, and evaluation criteria. Section 3 presents a detailed review of traditional path planning algorithms, examining their strengths, limitations, and suitable application contexts. Section 4 delves into the integration of machine learning and deep learning techniques in UAV path planning. Section 5 discusses energy-efficient path planning strategies, focusing on their role in optimizing flight endurance. Section 6 explores the unique challenges and methodologies of cooperative path planning for multiple UAVs. Finally, Section 7 outlines the ongoing challenges in UAV path planning research and provides a forward-looking perspective on potential future advancements in the field.

2. Fundamentals and Categorization of UAV Path Planning

2.1. Problem Definition

As a fundamental aspect of UAV applications, UAV path planning seeks to determine an optimal flight path that begins at the starting point and reaches the target point with precision. Assume the UAV's flight path is represented by a continuous function

$$\mathbf{P} : [0, T] \rightarrow \mathbb{R}^n,$$

which satisfies the boundary conditions

$$\mathbf{P}(0) = \mathbf{P}_s, \quad \mathbf{P}(T) = \mathbf{P}_t.$$

The goal of UAV path planning is to determine an optimal path that minimizes an overall cost function while meeting a set of constraints. In general, the cost function can be expressed as

$$J(\mathbf{P}) = \alpha L(\mathbf{P}) + \beta E(\mathbf{P}) + \gamma R(\mathbf{P}) + \dots,$$

where

- $\alpha, \beta, \gamma, \dots$ are weight coefficients for the respective objectives.
- $L(\mathbf{P})$ represents the path length:

$$L(\mathbf{P}) = \int_0^T \|\mathbf{P}'(t)\| dt,$$

which is crucial for minimizing flight time under a constant velocity assumption.

- $E(\mathbf{P})$ denotes the energy consumption cost:

$$E(\mathbf{P}) = \int_0^T \ell_E(\mathbf{P}(t), \mathbf{P}'(t), t) dt,$$

where ℓ_E is a function representing the instantaneous energy usage, influenced by factors such as flight speed, altitude, and air resistance.

- $R(\mathbf{P})$ is a cost function associated with collision risk or obstacle avoidance (e.g., penalizing the path when $\mathbf{P}(t)$ enters an unsafe region $\mathcal{O}$).

By “optimal”, we refer to the achievement of a predefined objective while adhering to a complex set of constraints. The various types of path planning can be broadly classified as shown in Table 1.

Table 1. Key objectives in UAV path planning.

Objective	Description
Minimum length path	Determines the route that minimizes total distance traveled, reducing flight time and enhancing task efficiency. Under constant-velocity conditions, this path corresponds to the shortest flight duration [44,45].
Energy consumption path	Focuses on minimizing UAV energy usage by considering factors such as flight speed, altitude changes, and air resistance. Rational path planning ensures optimal energy efficiency during the mission [30].
Collision-free path	Ensures UAV safety by navigating around static and dynamic obstacles. Hybrid navigation approaches allow safe operation in 3D, partially unknown, and dynamic environments [46].
Multi-objective optimization path	Balances multiple objectives, such as minimizing travel time, energy consumption, and risks. Advanced algorithms are used in complex scenarios like military reconnaissance or emergency rescue to achieve optimal trade-offs [47,48].

Constraints in UAV Path Planning

In the process of UAV path planning, various constraints must be incorporated to ensure the feasibility and safety of the mission. These constraints can arise from environmental factors, operational limitations, or specific mission requirements. The following elaborates on various types of constraints.

- **Environmental constraints:** As a crucial restrictive factor within the UAV flight environment, environmental constraints cover a wide and complex range of elements. Static obstacles, such as terrain and landforms in natural geographical environments, and buildings in populated areas, are key components of this constraint. Dynamic obstacles, whose positions and motion states continuously change, include other UAVs, birds, and moving vehicles. In recent years, flight areas defined by management regulations and safety requirements have also become part of environmental constraints. These include no-fly zones [49] and restricted-fly zones, such as ecological protection areas [50].
- **Physical constraints:** Physical constraints [51] arise from the UAV’s inherent characteristics and performance limitations, directly defining the feasible range of its flight path. The UAV’s hardware configurations, such as the power system and propulsion device [52], determine key performance metrics like flight speed and endurance range. Furthermore, due to structural design considerations aimed at ensuring flight stability, some UAVs have specific restrictive conditions during maneuvers, such as minimum turning radius and maximum turning angle [53]. Additionally, there are constraints related to the UAV’s flight capabilities, including maximum acceleration and maximum flight altitude [54].

- **Task constraints:** Task constraints are directly associated with the specific missions assigned to UAVs. These typically include factors such as the priority of flight objectives and task time. The priority constraint becomes particularly relevant in practical applications where it is crucial to prioritize the swift execution of high-priority tasks during path planning [55]. To meet the time-sensitive nature of these tasks, other objectives, such as minimizing energy consumption and reducing flight distance, may need to be compromised to some extent.
- **Energy reservation constraints:** Energy consumption during UAV flight is influenced by multiple factors, including flight speed, altitude changes, flight attitude adjustments, and air resistance [56–59], all of which have a particularly significant impact. For instance, ref. [60] presents an effective energy-aware path planning method that efficiently navigates through uncertain and time-varying wind fields, designing a path that leverages the energy provided by these wind conditions.

2.2. Categorization of UAV Path Planning Approaches

From an academic perspective, based on various classification criteria, UAV path planning problems can be categorized into the following types, as shown in Figure 3.

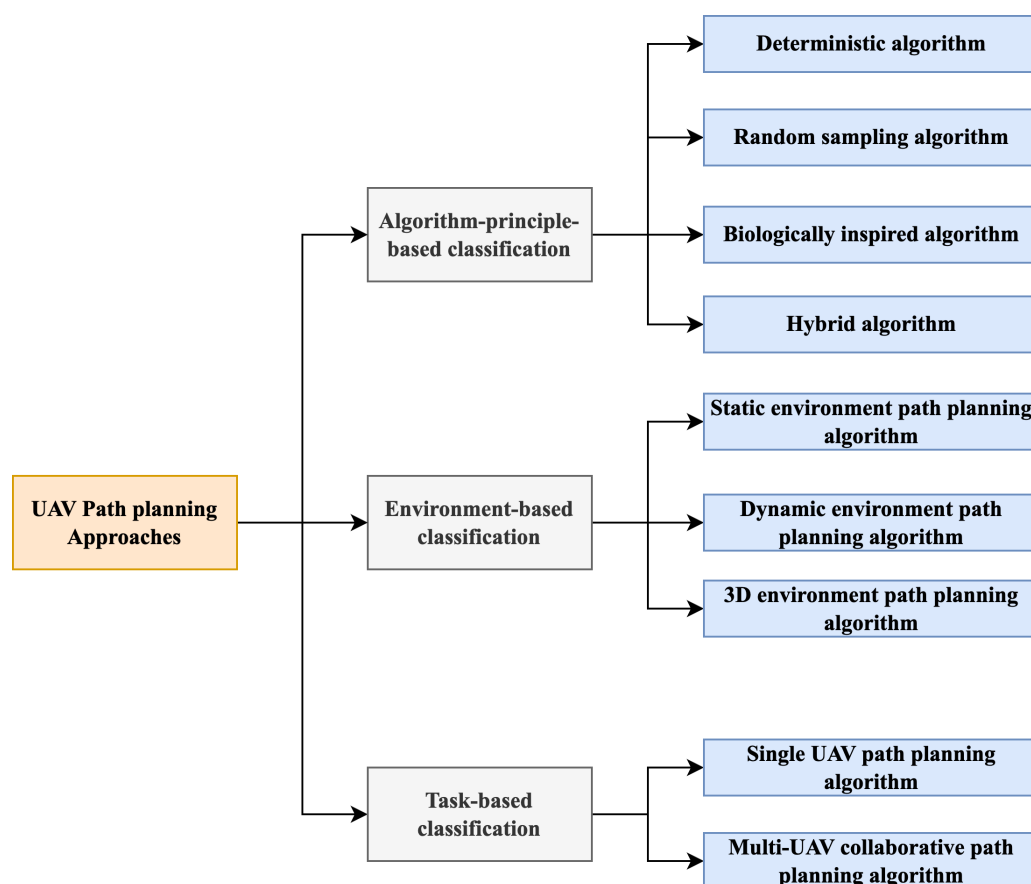


Figure 3. Classification of UAV path planning methods across diverse criteria.

2.2.1. Algorithmic Principle-Based Classification

- **Deterministic algorithms:** Deterministic algorithms follow fixed rules and steps, producing consistent results for the same input, making them predictable and stable. They are suitable for scenarios requiring high accuracy in path planning. Representative methods include graph search algorithms and optimization algorithms. In graph search algorithms, the Dijkstra algorithm finds the shortest path in a weighted graph using a greedy strategy and priority queue [61]. The A* algorithm combines

breadth-first search with Dijkstra, using an evaluation function to prioritize nodes [62]. Linear programming, as an optimization algorithm, transforms the path planning problem into a mathematical model to maximize or minimize a linear objective under constraints.

- **Random sampling algorithms:** Random sampling algorithms construct a search framework by randomly sampling the space, making them effective for high-dimensional or complex environments [63]. A typical example is the RRT algorithm [64], which grows a tree from the start point by adding new nodes toward random points if they are collision-free. Variants like RRT* [65] optimize the path with a rewiring mechanism, while RRT-Connect [66] uses bidirectional search for improved efficiency in large spaces.
- **Biologically inspired algorithms:** Bio-inspired algorithms mimic natural behaviors to solve path planning problems. Examples include genetic algorithms, which simulate heredity and natural selection to evolve solutions [67]; PSO, which models the group foraging behavior of birds or fish [68,69]; and ACO, which mimics ant pheromone-based pathfinding [70].
- **Hybrid algorithms:** Hybrid algorithms combine different methods to overcome individual limitations, achieving better efficiency and quality in path planning. For instance, a hybrid of PSO and genetic algorithms improves 3D UAV path planning [71]. RRT-based hybrid methods enhance time efficiency [72], while RT-Dijkstra integrates RRT with deterministic planning for cluttered environments [73].

2.2.2. Environment-Based Classification

- **Static environment path planning:** Static environment path planning generates collision-free paths in settings with stationary obstacles, assuming fixed positions and geometries. It computes an optimal path from a start to a target location and is widely used in indoor robotic navigation [74], warehouse logistics [75], and autonomous mobility systems [76]. Its key advantage is computational efficiency, as the derived path requires no real-time updates due to the static nature of the environment.
- **Dynamic environment path planning:** Dynamic environment path planning handles scenarios with moving obstacles, requiring continuous real-time adaptation to ensure collision-free navigation. Applications include autonomous vehicles avoiding other vehicles, pedestrians, or road obstructions [77]. Unlike static planning, it integrates both static and dynamic obstacle considerations, relying on real-time sensor data, predictive modeling, and adaptive decision making [78]. This approach is computationally more complex due to the need for continuous trajectory updates.
- **Three-dimensional environment path planning:** Three-dimensional environment path planning is essential for navigating vehicles or drones in volumetric spaces, incorporating dimensions like altitude and heading. It is critical for UAV navigation [76], spacecraft trajectory optimization [79], and navigating complex terrains like mountains or urban areas [80]. The complexity arises from modeling volumetric spaces, collision detection [81], and processing large-scale 3D data, requiring robust and scalable algorithms.

2.2.3. Task-Based Classification

- **Single-UAV path planning:** Single-UAV path planning focuses on determining an optimal route for a UAV from a specified starting point to a designated target. In practical applications, this includes scenarios such as planning the UAV's trajectory from a distribution center to a customer's location for express delivery [82] or defining the flight path from a take-off point to a mapping area for field surveys [83,84]. The mission

objectives of single-UAV path planning are diverse. One primary goal is to minimize the flight distance, thereby reducing travel time and optimizing energy consumption to extend flight endurance, which is particularly critical for long-duration missions. Additionally, path planning must account for risk factors such as adverse weather conditions, ensuring that the selected route minimizes potential hazards and enhances flight safety.

- **Multi-UAV collaborative path planning:** Multi-UAV collaborative path planning involves coordinating multiple UAVs to efficiently complete tasks within a shared environment [85]. This process addresses several challenges, including task allocation—ensuring that each UAV is assigned appropriate duties—and path coordination to prevent conflicts or collisions. Additionally, the approach focuses on overall optimization to enhance task efficiency and system reliability. Such collaborative planning is particularly critical in complex scenarios, including search and rescue operations [86] and environmental monitoring [87], where the collective performance of multiple UAVs significantly influences mission success.

2.3. Evaluation Metrics

In evaluating the performance of a path planning algorithm, a comprehensive assessment should be conducted based on several key indicators. At this stage, these indicators are defined and analyzed in terms of their influencing factors, as shown in Table 2.

Table 2. Evaluation metrics for UAV path planning algorithms.

Evaluation Metric	Description	Influencing Factors
Path length	The total length of the path, typically used to measure the brevity of the route, especially in shortest-path problems.	Obstacle distribution, environmental complexity
Computation time	The time required by the path planning algorithm. In real-time tasks, it directly determines whether the task can be completed on schedule.	Environmental complexity, algorithm efficiency, hardware performance
Energy consumption	The energy consumed during the UAV's flight, often related to battery usage. Optimizing energy consumption is critical for long-duration tasks.	Flight trajectory, altitude, speed control, environmental factors
Path safety	The safety of the path, assessing its ability to avoid collisions with static and dynamic obstacles to prevent UAV damage.	Obstacle types, density, dynamics
Path smoothness	The smoothness of the path, ensuring it aligns with the UAV's motion characteristics. Sharp turns may hinder efficient task execution.	Flight control limitations, smoothing strategies in the algorithm
Robustness	The adaptability of the algorithm to environmental changes and uncertainties, such as dynamic obstacles in changing environments.	Algorithm adaptability, real-time responsiveness

3. Classical Path Planning Algorithms

3.1. Graph-Based Algorithms

Graph-based path planning seeks to find the shortest path between a start node s and a goal node t in a weighted graph $G = (V, E)$. The objective is to minimize the total path cost:

$$\text{Cost}(P) = \sum_{(u,v) \in P} w(u,v),$$

where $P = \{s, v_1, v_2, \dots, t\}$ is the path. The cost function $d(v)$ represents the minimum cost to reach node v from s and is iteratively updated as

$$d(v) = \min(d(v), d(u) + w(u, v)), \quad \forall (u, v) \in E.$$

For graph-based methods (e.g., A^*), a heuristic function $h(v)$ estimates the cost from v to t , with the total cost given by

$$f(v) = d(v) + h(v).$$

The algorithm terminates when the goal node t is reached. The commonly used graph-based algorithms, including Dijkstra, A^* , and D^* methods, are described in Figure 4.

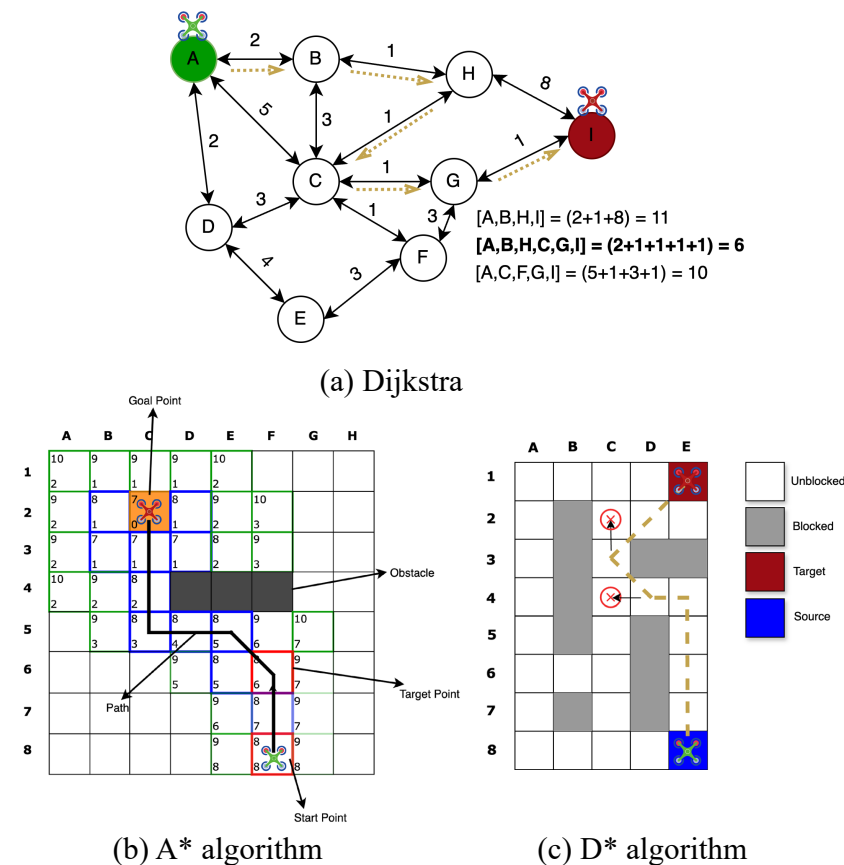


Figure 4. The principles of commonly used graph-based algorithms: Dijkstra, A^* , and D^* methods.

- The Dijkstra algorithm is a single-source shortest-path method based on weighted graphs. It iteratively updates the shortest distances from the source node to all other nodes until the target is reached, ensuring a globally optimal solution in static environments [88]. This algorithm performs exceptionally well in grid-based maps and road networks, making it suitable for applications such as airport patrol [89] and indoor navigation [90]. However, its high computational complexity and limited scalability in large or dynamic environments present significant challenges.
- The A^* algorithm enhances Dijkstra's approach by incorporating a heuristic function that estimates the cost from any node to the goal. By combining the actual cost and the heuristic estimate, A^* significantly accelerates the search for the optimal path. This algorithm is widely applied in static environments where both optimality and efficiency are required. However, in high-dimensional spaces or dynamic settings, designing an effective heuristic function becomes challenging, potentially limiting

performance. Variants such as Theta* [91] have been proposed to address issues like path smoothness in grid-based implementations.

- The D* algorithm is a dynamic path planning algorithm designed to adapt to environmental changes by incrementally updating the planned path. Once an initial path is established, D* leverages previously computed information to efficiently re-plan when obstacles or other environmental modifications are detected. The core principle involves adjusting the priority of nodes based on their cost and distance to the target while maintaining a priority queue. When the environment changes, only the affected nodes require recalculation, avoiding the need to recompute the entire path [92]. While D* excels in dynamic scenarios, frequent environmental changes can significantly increase computational overhead.

As seen above, graph-based algorithms have significant advantages in static low-dimensional environments. However, in high-dimensional (such as 3D urban spaces) or dynamic scenes, the inherent contradiction between their computational complexity and real-time performance becomes prominent: pursuing the optimal solution often comes at the expense of efficiency, and simplifying the model will reduce security. It is difficult to balance the two.

3.2. Optimization Methods

- **Linear programming (LP)** formulates the path planning problem as an optimization task governed by linear constraints and an objective function. By utilizing established LP solvers, this approach efficiently computes globally optimal solutions in environments characterized by simple constraints and static conditions. However, its effectiveness diminishes in scenarios with complex, nonlinear constraints or dynamic environments, where its assumptions and computational efficiency become limiting factors.
- **Nonlinear programming (NLP)** methods address path planning problems with complex constraints, such as dynamic obstacles and curved trajectories, that linear models cannot adequately represent. By incorporating nonlinear objective functions and constraints, NLP enables the generation of paths that more accurately capture the physical and operational requirements of the task. This approach is particularly advantageous in scenarios such as UAV formation control and continuous motion planning for robots. However, its practical application is often hindered by high computational complexity and susceptibility to local optima, posing significant challenges in real-time and large-scale environments.

The accuracy of optimization methods is based on accurate mathematical modeling. However, the strong nonlinearity and environmental uncertainty of UAV dynamics lead to a “modeling gap” between model simplification and actual constraints, which limits their application in highly dynamic scenarios.

3.3. Random Sampling Algorithms

Random sampling algorithms aim to find a feasible path from a start state q_{start} to a goal state q_{goal} in a configuration space $\mathcal{C}$, avoiding obstacles $\mathcal{C}_{\text{obs}} \subset \mathcal{C}$. These methods rely on sampling random configurations $q_{\text{rand}} \in \mathcal{C}_{\text{free}} = \mathcal{C} \setminus \mathcal{C}_{\text{obs}}$ to incrementally construct a graph $G = (V, E)$ or a tree T . The principles of random sampling algorithms, such as the Probabilistic Roadmap (PRM) and Rapidly Exploring Random Trees (RRT), are outlined as follows.

- **The Rapidly Exploring Random Trees (RRT)** algorithm incrementally constructs a search tree to explore the configuration space by randomly sampling nodes and connecting them to the nearest neighbors in the existing tree, as shown in Figure 5a,b.

Mathematically, the process can be described as follows: a random configuration x_{rand} is sampled from the free space $\mathcal{C}_{\text{free}}$, the nearest node x_{near} in the current tree T is identified based on a distance metric $x_{\text{near}} = \arg \min_{x \in T} \|x - x_{\text{rand}}\|$, and a new node x_{new} is generated by extending from x_{near} toward x_{rand} by a fixed step size e , such that $x_{\text{new}} = x_{\text{near}} + e \cdot \frac{x_{\text{rand}} - x_{\text{near}}}{\|x_{\text{rand}} - x_{\text{near}}\|}$. If the path from x_{near} to x_{new} lies entirely within $\mathcal{C}_{\text{free}}$, the new node is added to the tree. A key advantage of RRT is its probabilistic completeness, meaning that as the number of iterations k approaches infinity, the probability $P(k)$ of finding a solution converges to one [93]. The choice of the extension step size e is critical, as it affects both the exploration efficiency and path quality. For instance, setting e to 5% of the diagonal length of the environment has been shown to achieve a success rate of 96% in path planning for a 3D robotic arm [94]. Unlike traditional search algorithms that often suffer from the curse of dimensionality in high-dimensional spaces, RRT effectively mitigates this issue through its random sampling mechanism, resulting in superior search efficiency. However, the algorithm has notable limitations: the stochastic nature of the search tree often leads to suboptimal paths, and it cannot guarantee a globally optimal solution. To address these issues, various improved algorithms have been proposed, as summarized in Table 3. Despite its limitations, the RRT algorithm remains a cornerstone of motion planning, offering a powerful and efficient approach for exploring large and complex configuration spaces.

- **The Probabilistic Roadmap (PRM) algorithm** constructs a collision-free path in high-dimensional configuration spaces by building a graph (roadmap) of sampled configurations and then searching for paths within this structure, as shown in Figure 5c,d. Mathematically, the process can be described in two phases: during the learning phase, random configurations $x_{\text{rand}} \in \mathcal{C}_{\text{free}}$ are sampled from the free space, and edges are created by connecting x_{rand} to its neighboring nodes x_{near} within a specified radius r , where a connection is valid only if the direct path between the nodes lies entirely within $\mathcal{C}_{\text{free}}$. In the query phase, graph search algorithms such as A* or Dijkstra are employed to find the shortest path between the start and goal nodes on the constructed roadmap. A key strength of PRM is its probabilistic completeness: as the number of samples n increases, the probability of finding a solution (if one exists) approaches one [95]. The algorithm's performance is heavily influenced by parameters like the sampling density n and connection radius r . For example, in a six-degrees-of-freedom (DOF) robotic arm scenario, using $n = 1000$ samples and $r = 0.2$ achieves a 92% success rate in cluttered environments [96]. PRM is particularly effective in high-dimensional spaces due to its decoupling of roadmap construction from path queries, enabling the reuse of the roadmap for multiple planning tasks, which contrasts with incremental methods like RRT that rebuild the tree for each query. However, PRM's preprocessing phase can be computationally expensive in complex environments, especially in the presence of narrow passages, as uniform sampling may fail to capture these critical regions. Moreover, the quality of the roadmap depends on the sampling strategy, and the algorithm does not guarantee optimal paths. Variants such as Lazy PRM [97] and PRM* [63] address these issues by optimizing sampling and connection strategies, improving path quality and computational efficiency. Despite its limitations, PRM remains a versatile and powerful tool for motion planning, particularly in scenarios that require repeated queries in static environments.

Random sampling algorithms are necessary in unknown environments, but their non-optimality and computational overhead mean they need to be combined with other algorithms in scenarios where the task is sensitive to the path length or the real-time requirement is extremely high.

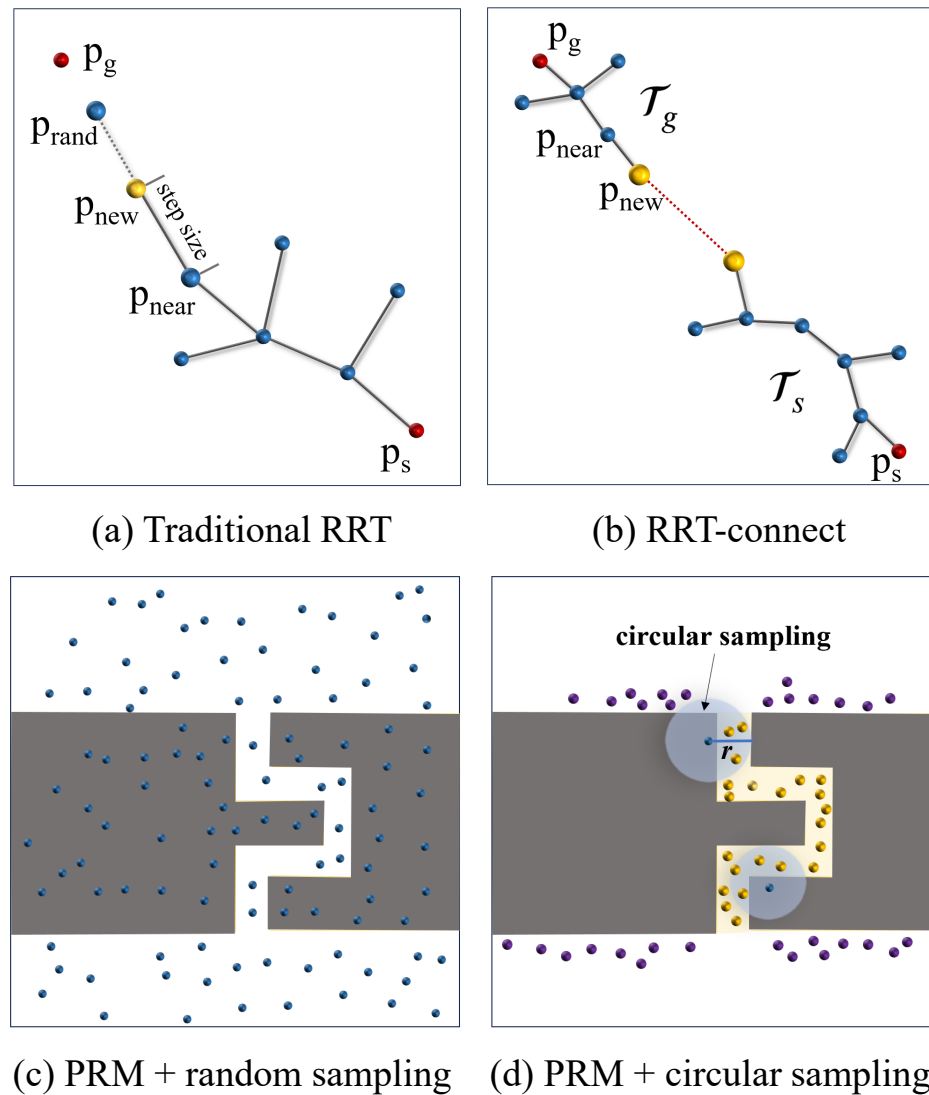


Figure 5. Different sampling strategy in RRT and PRM algorithms.

Table 3. Comparison of different RRT algorithms.

Algorithm	Improvement	Advantages	Reference
RRT*	Based on RRT; path optimization is introduced to ensure asymptotic optimization	Asymptotic optimality; better path quality	[65]
RT-RRT*	Adds tree maintenance and rewiring mechanisms	Efficient path generation; adaptable	[98]
Informed RRT*	Using a heuristic function to limit the search space	Less unnecessary exploration; faster	[99]
RRT*-smart	Uses intelligent sampling methods, incremental optimization, and heuristic guidance	Efficient path quality; strong adaptability	[100]
Bi-RRT*	The combination of retrospective thought and greedy search strategy; path optimization strategy and path smoothing strategy	High algorithm efficiency; high path quality	[101]
RRT-Connect	Improves the search speed by means of bidirectional tree	Bidirectional search; fast response	[66]

Table 3. Cont.

Algorithm	Improvement	Advantages	Reference
RRT-blossem	Using biased sampling. Local optimality is avoided by constraint equations	Overall search speed is fast	[102]
Circle-RRT	Sampling strategy combined with circular constraints	Circular constraint sampling; energy obstacle optimization.	[103]
KD-RRT	Test case generation strategy; nearest neighbor search optimization	Strong versatility; scalable; high efficiency	[104]
Dynamic RRT (DRRT)	Supports path replanning for dynamic obstacle update	Real-time replanning; dynamic obstacle adaptation.	[105]
Hybrid RRT (HRRT)	The RRT and A* algorithms are combined and a heuristic search strategy is used	Prefers target bias node; fast response	[106]

3.4. Bio-Inspired Algorithms

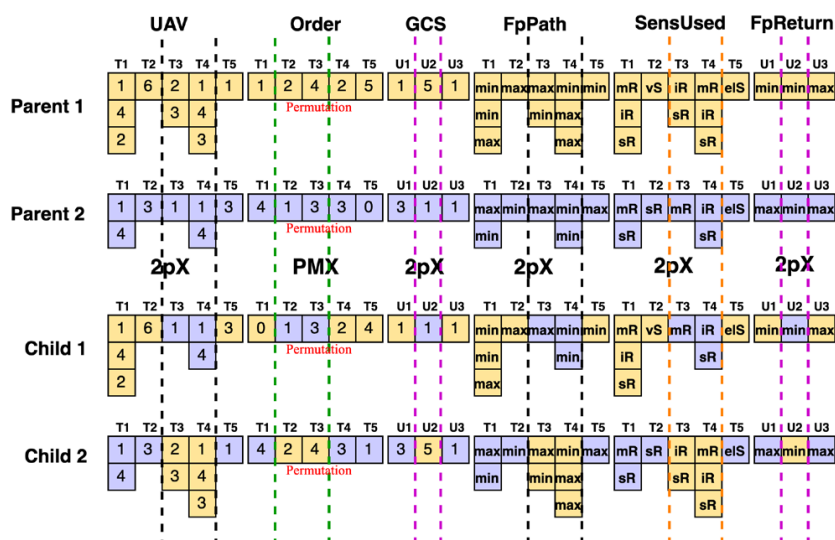
Bio-inspired algorithms are a class of computational techniques inspired by natural phenomena and biological processes, designed to solve complex optimization and search problems. Genetic algorithms (GAs) mimic evolution through selection, crossover, and mutation. Particle swarm optimization (PSO) simulates the collective behavior of swarms, like birds or fish, to find optimal solutions. Ant colony optimization (ACO) models ant foraging behavior using pheromone trails to solve pathfinding problems. Hybrid algorithms combine multiple methods to improve efficiency and adaptability for complex tasks. Specific explanations are as follows.

- **Genetic algorithms (GAs)** simulate the process of natural evolution, treating candidate paths as individuals that evolve over successive generations through selection, crossover, and mutation operations [107]. GAs are particularly effective in exploring complex, multi-objective optimization landscapes, making them a popular choice for multi-UAV cooperative tasks [108] and other scenarios with intricate constraints, as shown in Figure 6a. However, GAs are susceptible to premature convergence to local optima and often exhibit slower convergence rates. This necessitates careful parameter tuning and, in many cases, hybridization with other methods to enhance performance.
- **Particle swarm optimization (PSO)** simulates the social behavior observed in flocks of birds or schools of fish searching for food, where each particle in the swarm represents a potential solution and adjusts its trajectory based on both individual and collective best experiences, as shown in Figure 6b. Known for its rapid convergence and simplicity in parameter configuration, PSO is particularly attractive for static environments that require energy-efficient path planning. However, its performance can be highly sensitive to the initial distribution of particles, and its ability to adapt to dynamic changes is generally limited [109,110].
- **Ant colony optimization** models the foraging behavior of ants, where artificial agents (ants) deposit virtual pheromones along the paths they traverse, as shown in Figure 6c. Over time, paths with higher pheromone concentrations become more attractive, guiding the swarm toward an optimal solution [111]. ACO is robust and adaptable, making it particularly useful in scenarios involving dynamic task allocation and complex route planning. However, the algorithm often requires extensive computation and iterative processing, which leads to slower convergence and increased computational complexity, especially in large-scale problems [112,113].
- **Hybrid algorithms** have emerged as a promising strategy to overcome the inherent limitations of individual methods [114]. By integrating multiple techniques, hybrid algorithms capitalize on their complementary strengths, effectively addressing the

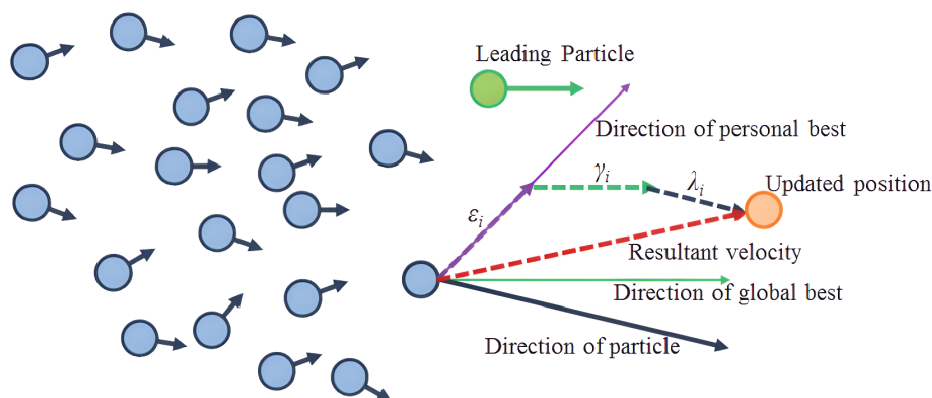
challenges posed by dynamic environments and complex constraints. Path planning algorithms are typically defined by distinct performance characteristics and limitations [115]. For example, deterministic methods such as A* provide optimal solutions in static environments but lack the flexibility needed to adapt to real-time changes. In contrast, sampling-based algorithms like RRT excel in quickly exploring high-dimensional spaces, but may generate suboptimal or non-smooth paths. Bio-inspired methods, including genetic algorithms (GAs) and particle swarm optimization (PSO), offer robust global search capabilities, yet are susceptible to challenges such as premature convergence and sensitivity to parameter initialization. The integration of these diverse approaches is motivated by the need to harness their respective strengths—achieving global optimality while ensuring rapid local adaptability—thereby facilitating more robust and efficient performance in both static and dynamic operational contexts.

Metaheuristic algorithms [116], inspired by natural phenomena, such as evolution, swarm intelligence, and animal behavior, have emerged as powerful tools for addressing the challenges of UAV path planning in real-world environments. Traditional path planning methods often struggle to cope with dynamic and unpredictable obstacles, sensor noise, and the need for real-time decision making. In contrast, metaheuristic approaches—such as particle swarm optimization (PSO), ant colony optimization (ACO), genetic algorithms (GAs), and more recently, nature-inspired techniques like the intelligent Beetle Antennae Search (BAS) [117]—offer adaptive and flexible frameworks that can significantly improve the robustness of UAV path planning. For instance, the BAS algorithm [118] emulates the foraging behavior of beetles, using virtual “antennae” to sense and avoid obstacles in real time. By continuously evaluating the environment and updating the UAV’s trajectory based on local and global information, such algorithms enable UAVs to efficiently navigate complex and cluttered spaces while minimizing the risk of collision. Furthermore, metaheuristics are well suited to handling multi-objective optimization, allowing UAVs to balance competing requirements such as energy efficiency, flight time, and safety. Another key advantage of metaheuristic algorithms is their ability to escape local optima and adapt to changes in the environment, which is essential for robust operation in dynamic settings. By integrating these algorithms into UAV path planning systems, it is possible to achieve greater resilience against uncertainties and unexpected events, such as sudden appearance of obstacles or changes in mission objectives. Overall, the use of metaheuristic techniques not only enhances the adaptability and reliability of UAV navigation but also paves the way for more intelligent and autonomous aerial systems capable of operating safely in real-world scenarios.

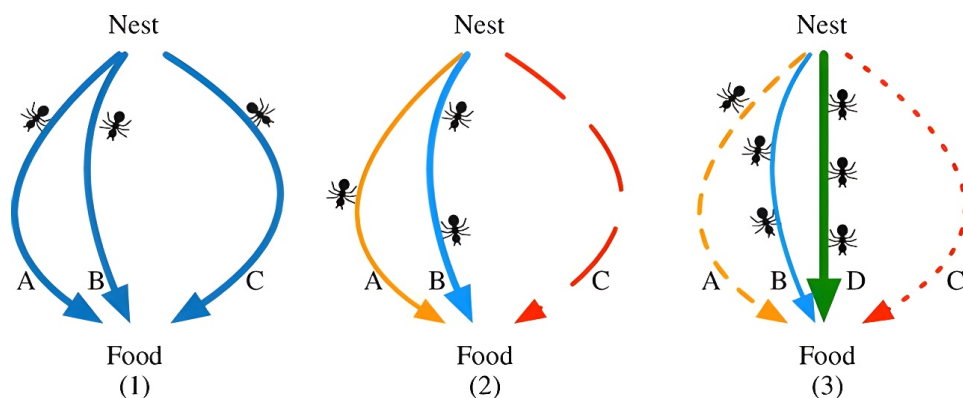
Model predictive control (MPC) [119] has become a widely adopted technique in path planning for autonomous systems due to its ability to handle multi-constraint optimization problems in real time. By predicting future system states over a finite time horizon and optimizing control inputs accordingly, MPC enables vehicles to navigate complex and dynamic environments while satisfying safety and performance constraints. In addition, sampling-based optimal control methods, such as Rapidly Exploring Random Trees (RRT*) and Monte Carlo-based approaches, have gained attention for their effectiveness in high-dimensional or non-convex spaces. These algorithms generate feasible trajectories by sampling the state space and iteratively seeking optimal solutions, making them particularly suitable for applications where traditional grid-based methods struggle with scalability and computational efficiency.



(a) Genetic algorithm



(b) Particle swarm optimization



(c) Ant colony optimization

Figure 6. The principles of bio-inspired algorithms, encompassing genetic algorithms, particle swarm optimization, and ant colony optimization.

Game theory [120] provides a powerful framework for addressing the challenges of multi-agent path planning, especially in scenarios involving cooperation, competition, or negotiation among autonomous agents. By modeling the interactions between agents as strategic games, researchers can design algorithms that account for the objectives and

constraints of each participant, leading to more robust and adaptive path planning solutions. Approaches such as Nash equilibrium-based planning, Stackelberg games, and cooperative bargaining have been employed to resolve conflicts, promote coordination, and ensure efficient resource allocation among multiple vehicles. These methods are particularly relevant for scenarios like UAV–UGV cooperation, swarm robotics, and autonomous traffic management, where the actions of one agent can significantly influence the outcomes of others.

3.5. Key Trade-Off Relationships Between Traditional Path Planning Algorithms

In order to compare the advantages and disadvantages of each algorithm more intuitively and systematically, we introduce Table 4 to conduct a horizontal comparison and analysis of the algorithms introduced in this section based on multiple key performance indicators such as search speed, obstacle avoidance ability, and path optimality. It can be seen that the Dijkstra algorithm has a fast search speed in simple graph structures, can find the shortest path between two points, and has good obstacle avoidance ability when the graph model is reasonably constructed. However, it has poor real-time performance in complex graph scenarios and struggles to cope with dynamic environments. When the heuristic function is good, the A* algorithm has a fast search speed and can also find the shortest path. The rest of its performance is similar to that of the Dijkstra algorithm. The Rapidly Exploring Random Trees algorithm stands out due to its fast search ability and adaptability to dynamic environments. However, its path is not optimal and can only achieve local path exploration. The particle swarm optimization algorithm has a fast convergence speed and relatively low computing cost. It performs well in terms of real-time performance and adaptation to dynamic environments, but it cannot guarantee the optimal path. The ant colony optimization algorithm achieves obstacle avoidance and adaptation to dynamic environments through pheromone mechanisms. However, it has high computational overhead and performs poorly in search speed and local path planning. Although the GA algorithm has strong global search ability and high maturity, it has problems such as slow convergence speed, high computing cost, and difficulty in real-time adjustment. Furthermore, through the comparison results, the advantages and limitations of each algorithm on different performance indicators can be revealed, providing a clear reference for the reasonable selection of algorithms in practical applications.

To further elucidate the technical characteristics and applicable boundaries of different path planning algorithms, Table 4 categorizes the four major types of algorithms into sub-categories and performs comparative analyses based on core principles, typical scenarios, advantages, and disadvantages. Additionally, Table 5 evaluates the performance of these algorithms by leveraging the content from Table 2, utilizing a ‘high, medium, low’ scale to compare the performance of each algorithm across various performance indicators. In practical applications, the selection of algorithms necessitates a comprehensive consideration of multi-dimensional factors. For instance, while the RRT algorithm exhibits strong real-time performance and adaptability to dynamic environments, it incurs relatively higher path lengths and energy consumption. The A* algorithm achieves an optimal balance between path length and security in static environments. Through this hierarchical and multi-dimensional analysis, researchers can efficiently identify suitable algorithms tailored to the requirements of specific tasks.

Table 4. Categorization of traditional path planning algorithms and core characteristics.

Category	Subdivided Algorithms	Core Principle	Typical Application Scenarios	Advantages	Disadvantages
Graph-Based Algorithms	Static Optimal Search (Dijkstra, A*)	Discrete graph model and priority queue; Dijkstra without heuristics, A* with heuristic function $h(v)$	Indoor navigation; road network path planning; grid map scenarios	Dijkstra: Globally optimal, theoretically complete; A*: Heuristic acceleration balancing efficiency and optimality; high path smoothness	Dijkstra: Complexity $O(V^2)$, inefficient for large-scale scenarios; A*: Heuristic-dependent, suboptimal with poor design; only suitable for known environments
	Dynamic Replanning (D*, D*-Lite)	Incremental replanning by reverse-updating affected nodes	Dynamic obstacle scenarios (e.g., disaster relief, dynamic obstacle avoidance)	Incremental update using historical information, avoiding global replanning; strong adaptability to dynamic environments	Dramatic overhead with frequent changes; time-consuming node priority calculation in complex environments
Optimization Methods	Linear Programming (LP)	Linear objective function and linear constraints, relying on mature solvers	Simple kinematic constraints	Fast solving, suitable for simple scenarios with low real-time requirements; mathematically guaranteed solution feasibility	Unable to handle nonlinear constraints; model simplification may lead to practical infeasibility
	Nonlinear Programming (NLP, SNLP)	Nonlinear objectives/constraints, supporting curved trajectories and kinematic limitations	UAV formation control 3D complex trajectory generation	Accurate modeling of UAV kinematic constraints; support for curved trajectory optimization adapting to complex kinematic limitations	High computational complexity, poor real-time performance; prone to local optima, dependent on initial point selection
Random Sampling Algorithms	Random Sampling Algorithms (RRT, RRT*, RRT-Connect)	Random sampling + tree structure expansion; RRT* for asymptotic optimality, RRT-Connect for bidirectional search acceleration	Unknown dynamic environments	No prior environment knowledge, suitable for unknown/dynamic scenarios; RRT* for asymptotic optimality, RRT-Connect for accelerated search	Poor path quality of basic RRT (non-optimal); low passage rate in narrow channels due to sampling randomness
	Probabilistic Roadmap (PRM, PRM*, Lazy PRM)	Preprocessing to construct collision-free graphs, using graph-based search for queries	Static high-dimensional environments	Efficient multiple queries after preprocessing; PRM* improves success rate in narrow areas via adaptive sampling	Long preprocessing time, unsuitable for dynamic environments; uniform sampling may miss critical passages
Bio-inspired Algorithms	GA	Evolution of path populations via selection, crossover, and mutation, supporting multi-objective optimization	Multi-UAV task allocation, complex constraint scenarios	Native support for multi-objective optimization; strong robustness adapting to complex constraint combinations; strong solution-space-exploration ability; prone to premature convergence into local optima;	Time-consuming evolution process, sensitive to parameters (crossover/mutation probabilities)
	PSO	Simulating bird/fish swarm behavior, iteratively updating individual and global best solutions	Rapid feasible solution generation	Fast convergence; few parameters, simple implementation	Particle aggregation in high-dimensional spaces, reduced search accuracy; dependent on initial distribution, weak global search capability
	ACO	Pheromone accumulation guiding path search, adapting to dynamic task allocation	Multi-robot collaboration, logistics distribution path optimization	Suitable for dynamic task allocation and path optimization; natural balance between exploration and exploitation via pheromone mechanism	High computational complexity, inefficient for large-scale scenarios; difficult parameter tuning for pheromone evaporation, prone to stagnation

Table 5. Qualitative analysis for path planning algorithms.

Dimension	Path Length	Consumption Time	Energy Consumption	Path Safety	Path Smoothness	Robustness	Multi-Objective Handling
Static Optimal Search	Theoretically shortest	High	Relatively low	High	High	High	Single objective
Dynamic Replanning	Close to optimal	Medium	Relatively low	Medium	High	Medium	Single objective and dynamic weights
LP	Theoretically optimal	Medium	Relatively low	Medium	High	Low	Single objective
NLP	Locally optimal solution length	Extremely high	Relatively low	High	High	Low	Single objective + weighted sum
RRT	Usually longer	Low	Relatively high	Medium	Low	Medium	Single objective
PRM	Probabilistically close to optimal	High	Medium	Medium	Medium	Low	Single objective
GA	Approaching optimality through iteration	High	Medium	High	Medium	High	Native support
PSO	Locally optimal solution length	Low	Medium	Medium	Medium	Medium	Weighted transformation to single objective
ACO	Asymptotically approaching optimal length	High	Medium	Medium	Medium	High	Multi-stage optimization

4. Advanced Artificial Intelligence Path Planning Methods

Artificial intelligence methods have revolutionized UAV path planning, enabling efficient navigation in complex and dynamic environments, which can be categorized into the following types, as shown in Figure 7. By leveraging techniques such as machine learning, reinforcement learning, and deep learning, innovative algorithms have been developed to optimize trajectories, enhance obstacle avoidance, and meet real-time computational demands.

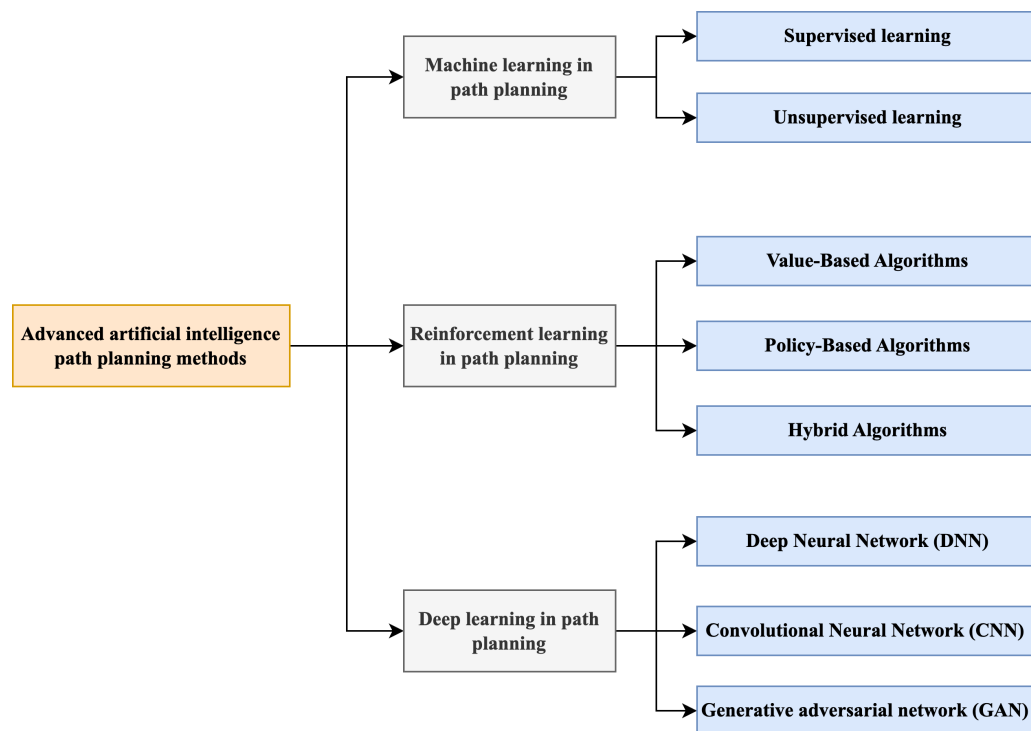


Figure 7. Classification of advanced artificial intelligence path planning methods.

4.1. Machine Learning in Path Planning

Machine learning (ML) has brought remarkable advancements to UAV path planning, especially when operating in unpredictable or rapidly changing scenarios. For example, Cui et al. [121] developed a path planning strategy for UAVs utilizing a two-layer reinforcement learning architecture, where separate Q-learning modules manage both global and local data streams. Their approach incorporates B-spline curves to generate smoother trajectories, which in turn boosts the effectiveness of the planning process. In another study, Afifi et al. [122] investigated how current 5G networks can be leveraged for autonomous UAV route planning, applying a combination of reinforcement learning and deep supervised learning methods. This blended technique not only tackles the inherent difficulties of path planning but also satisfies the stringent real-time processing needs of UAV missions. Qu et al. [123] introduced the RLGWO algorithm, merging reinforcement learning with gray wolf optimization to recast the path planning challenge as an optimization problem. Through adaptive switching that responds to reinforcement learning's cumulative outcomes, their solution adeptly manages UAV navigation in intricate and ever-changing environments. Taking a different route, Wang et al. [124] addressed navigation in obstacle-rich settings by proposing a method that compresses three-dimensional space using deep reinforcement learning. This innovation enables smarter trajectory choices while reducing computational demands, making it well suited for dynamic conditions. Furthermore, the AgilePilot platform employs deep reinforcement learning together with object detection, empowering UAVs to avoid moving hazards in real time. Sanyal et al. [125] presented

a Physics-Guided Neural Network (PgNN) model for autonomous UAV routing, which integrates object tracking and predicts future paths. Designed to lower energy usage, this model maintains safe navigation without collisions. By examining large-scale datasets, ML-based systems can uncover patterns and regularities, equipping UAVs to adjust rapidly to environmental shifts and demonstrating substantial adaptability and performance [126]. Core machine learning strategies used in UAV path planning encompass supervised learning, unsupervised learning, and reinforcement learning, each offering distinct strengths tailored to varying operational demands and mission objectives.

Supervised learning leverages labeled datasets to train predictive models, as depicted in Figure 8. The central objective is to learn a mapping from inputs to outputs, enabling the system to accurately predict outcomes for new, unseen data [127,128]. In real-world UAV applications, supervised learning can utilize historical flight records—such as chosen routes, velocities, and weather data—to develop regression or classification models that assist with flight path prediction. Regression techniques are well suited for forecasting continuous trajectories, while classification approaches can help select the most appropriate route from a set of predefined paths. For example, algorithms like support vector machines (SVMs) and decision trees are often applied to path classification, allowing the UAV to choose the optimal route based on current environmental factors [129]. Chen et al. [130] introduced an innovative UAV path planning framework using SVMs, demonstrating through experiments that this technique can produce both smooth and safe trajectories, even when flying near obstacles. In multi-UAV contexts, Gaussian processes (GPs) and their variants are widely adopted for route planning. Moon et al. [131] presented a cooperative search strategy for UAVs tracking moving targets, employing Gaussian process regression (GPR) to estimate target density and dynamically update reward functions for path planning, thus steering UAVs toward regions with higher target likelihood. Similarly, Zhang et al. [132] designed a multi-UAV tracking platform based on the Iterative Linear Quadratic Gaussian (iLQG) algorithm, systematically comparing methods for monitoring non-cooperative targets and optimizing both tracking precision and fuel consumption. Nevertheless, supervised learning comes with notable limitations, chiefly the dependence on high-quality labeled data. In complex or rapidly changing environments, collecting and annotating such datasets can be particularly challenging, which may restrict the practical effectiveness of supervised models.

Although machine learning methods perform well in term of dynamic environment adaptability, their effectiveness is highly dependent on the quality of the training data and the hyperparameter settings of the model. For example, in the multi-layer Q-learning method proposed by Cui et al. [130], the learning rate of the local Q-table is set to 0.01, and the exploration factor (ϵ) is initially set to 0.9 and gradually decays with training. This setting directly affects the speed of local path optimization. If the learning rate is too high, it is easy to cause training shock. If it is too low, the convergence speed will be slow. Furthermore, the path planning system combined with 5G infrastructure proposed by Afifi et al. [122] performed well in simulation experiments. However, in actual deployment, the quality of 5G signals declined due to environmental changes (such as weather and occlusion), which in turn affected the accuracy of the model's decision making and exposed the problem of unstable feature input. This suggests that the traditional ML model has insufficient robustness when facing the input distribution shift and needs to be improved by combining it with a domain adaptation method. In the RLGWO method proposed by Qu et al. [123], the hyperparameters of the gray wolf optimizer (GWO), such as the step size attenuation coefficient (a), have a significant impact on the convergence speed and the global optimal search ability. If the step size decays too quickly, it causes the algorithm to fall into local optimum. Although RLGWO enhances path diversity, when the

number of target points increases sharply, the algorithm complexity rises and the planning time increases significantly, which limits its real-time application potential in large-scale dynamic scenarios. Overall, machine learning methods can effectively model static or medium–low-dynamic environments. However, in rapidly changing and highly complex tasks, there are still problems such as poor generalization ability, high data dependence, and high hyperparameter debugging costs. In contrast, reinforcement learning and deep learning have greater potential in adaptability and feature extraction, but they also bring new challenges.

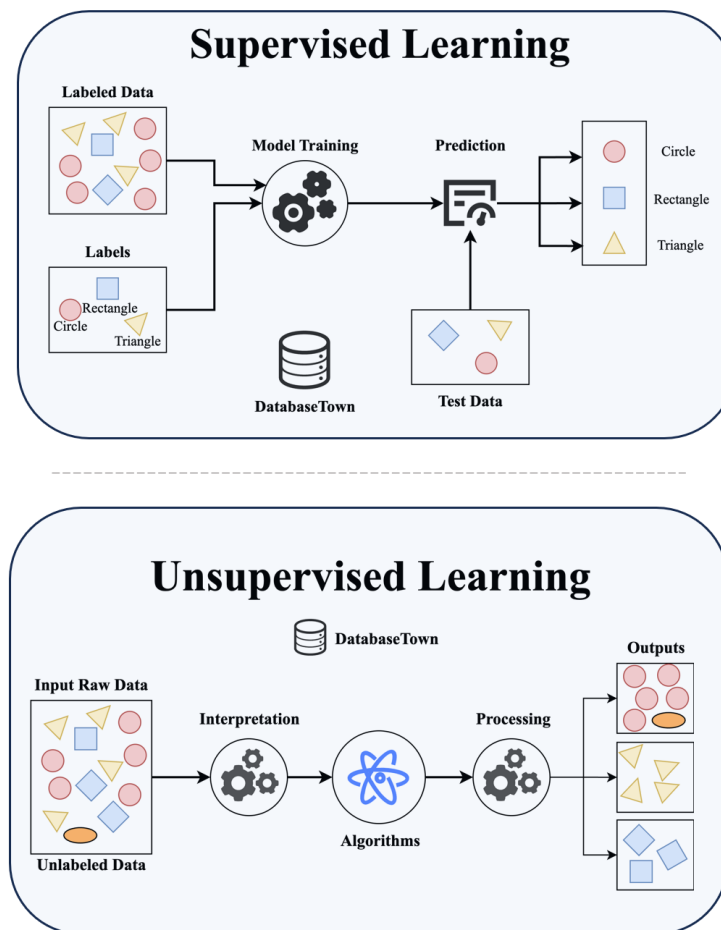


Figure 8. The theories of supervised learning and unsupervised learning.

4.2. Reinforcement Learning in Path Planning

Reinforcement learning (RL) enhances decision making by allowing agents to refine their actions based on feedback, typically in the form of rewards. In the context of path planning, RL plays a vital role in enabling UAVs to operate effectively within unpredictable and changing environments, addressing issues such as real-time obstacle avoidance [133] and optimizing routes for energy saving [134]. Through carefully designed reward mechanisms, RL empowers UAVs to autonomously steer clear of hazards, shorten travel times, and conserve power. Li et al. [135] improved the Deep Deterministic Policy Gradient (DDPG) algorithm for UAV ground target pursuit by refining reward structures, fostering cooperation among multiple UAVs, and integrating long short-term memory (LSTM) networks. Their approach surpassed conventional DDPG, achieving a 91.8% tracking rate in sparse settings and 67.5% in denser areas, along with faster convergence and greater training stability. He et al. [136] introduced a transparent deep reinforcement learning (DRL) framework for small-UAV navigation, validating its effectiveness through both simulated and real-world experiments and confirming its practicality and robust performance. Zhang et al. [137]

presented the SD-TD3 algorithm, which tackles the challenge of sparse rewards in RL by combining state detection coding with dynamic reward strategies, thereby improving both obstacle avoidance and convergence in complex scenarios. Wang et al. [124] created a DRL-based technique that leverages three-dimensional spatial information compression (3DSIC) to boost planning efficiency in 3D environments. By collapsing 3D data into two dimensions, and incorporating a Deep Q-Network (DQN), as illustrated in Figure 9, their method accelerated training by factors of 4.028 and 3.9, significantly enhancing UAV navigation performance. Despite these notable advances, RL remains challenged by factors such as unstable training processes, considerable computational demands, and the complexity of modeling high-dimensional state spaces, all of which can hinder overall efficiency and effectiveness.

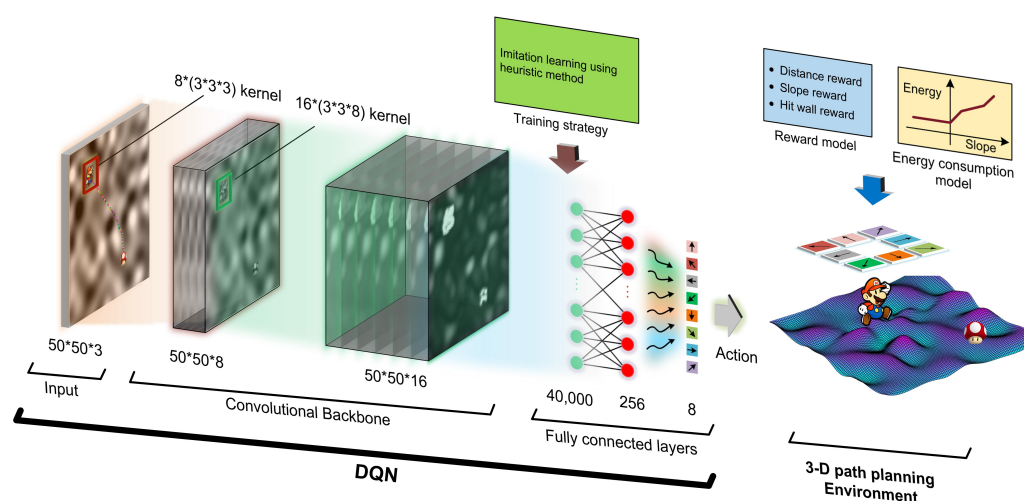


Figure 9. Deep Q-Network (DQN) for 2.5D path planning task [138].

Reinforcement learning (RL) empowers UAVs to discover optimal navigation routes through iterative trial-and-error processes, making it highly effective in managing the complexities of dynamic and unpredictable environments [139,140]. Yan et al. [141] improved Q-learning by crafting specialized reward functions, refining action selection mechanisms, and enhancing the initialization of Q-values, which collectively led to significant performance gains in adversarial conditions, as demonstrated using the STAGE Scenario platform. Yang et al. [142] utilized Deep Q-Networks (DQNs) for multi-robot path planning, integrating prior domain knowledge and rule-based enhancements to accelerate convergence and boost learning efficiency, achieving promising outcomes in warehouse scheduling tasks. Yan et al. [143] further advanced DQNs by introducing Dueling Double Deep Q-Networks (D3QN), resulting in more accurate Q-value estimation and the generation of safer, higher-reward paths in both static and dynamic settings. Tian et al. [144] and Bouhamed et al. [145] adopted the Deep Deterministic Policy Gradient (DDPG) algorithm for UAV guidance. Tian et al. incorporated hierarchical reinforcement learning along with a three-step experience replay buffer, significantly enhancing convergence speed and path accuracy in urban 3D environments. In contrast, Bouhamed et al. showcased effective obstacle avoidance in open 3D spaces by applying transfer learning strategies and meticulously engineered reward structures. As illustrated in Figure 9, RL methods such as Q-learning, DQN, and policy gradient algorithms serve as foundational tools for UAV path planning, enabling solutions to intricate challenges encountered in real-world scenarios.

Reinforcement learning demonstrates excellent autonomous decision-making ability in the path planning of unmanned aerial vehicles (UAVs), but its performance is highly dependent on the design of the reward functions, the selection of network architectures, and the adjustment of the hyperparameters. For example, in the improved DDPG proposed

by Li et al. [135], the actor network uses a two-layer fully connected network, while the critic network fuses the state and action inputs and then undergoes two-layer processing. This architecture effectively avoids the overfitting of the Q function, but it also brings higher computational overhead. In terms of hyperparameters, the key parameters of DDPG include the learning rate, discount factor γ , soft update parameter τ , and playback buffer size. These parameters have a direct impact on the training stability and convergence speed. If the learning rate is set too large, it is easy to cause strategy oscillation. If the playback buffer is insufficient and the sample diversity is poor, it leads to overfitting. In practical applications, there are typical failure cases of reinforcement learning methods. For example, the SD-TD3 algorithm proposed by Zhang et al. [137] is prone to the problem of sparse rewards in the early stage of training, resulting in slow policy learning. Furthermore, in highly dynamic environments (such as urban air traffic management), traditional RL methods often have difficulty adjusting immediately when facing unseen obstacle distributions, exposing the problem of insufficient generalization ability. To address this issue, the interpretable deep reinforcement learning (DRL) method proposed by He et al. [136] enhances the quantitative analysis of the importance of input features by introducing a model interpretation framework based on feature attribution. This method helps identify the influence of key input features on the strategy output by generating visual saliency maps and text interpretations. However, in the simulation experiments of complex urban environments, this method still failed to avoid obstacles under extreme obstacle layouts, indicating that there is still room for improvement in the aspect of security guarantees in reinforcement learning. Although reinforcement learning can obtain highly adaptive path planning strategies through training, its training cost is huge, and the model is prone to performance degradation (reality gap) in the actual environment after deployment. Especially on resource-constrained unmanned aerial vehicle (UAV) platforms, when deploying models with a large number of parameters such as DQN and DDPG, there exist problems of inference delay and energy consumption.

4.3. Deep Learning in Path Planning

Deep learning (DL) leverages multi-layer neural networks to automatically extract features, making it especially effective for processing large-scale, high-dimensional, and complex datasets [146]. For instance, Pan et al. [147] introduced a deep learning algorithm trained via genetic algorithms (DL-GA) for multi-UAV data collection path planning. Their method utilizes genetic algorithms to accumulate path planning experience, which is then used to train deep neural networks. Their experimental results indicate that DL-GA achieves significantly faster solution times compared to traditional genetic algorithms, particularly excelling as the number of data nodes increases. Bayerlein et al. [148] developed a multi-agent reinforcement learning framework for optimizing the paths of multiple UAVs, collecting data from distributed IoT devices. By modeling the problem as a decentralized partially observable Markov decision process (Dec-POMDP) and employing map-based processing alongside deep reinforcement learning, their framework produces adaptable policies that maintain robust performance across a range of environmental conditions. Additionally, deep learning has been integrated with classic path planning algorithms in related research. For example, Gabriel G. R. de Castro et al. [149] proposed an online adaptive method that combines fast extended random tree (RRT) algorithms with deep reinforcement learning (DRL) for UAV-based olive tree pest trap inspections. This hybrid approach enhances both reliability in path planning and obstacle avoidance for drones. In UAV path planning, deep learning technologies are primarily employed to extract informative features from complex environmental inputs—such as visual data, LiDAR, and other sensors [150–152]—and to generate effective routes based on these learned representations.

4.3.1. Deep Neural Network (DNN)

Deep neural networks (DNNs) play a crucial role in path planning by extracting complex environmental features through multi-layer nonlinear transformations, as illustrated in Figure 10. By processing environment-aware data, DNNs can generate adaptive routes tailored to various surroundings. In practice, these models have been applied to path prediction tasks across diverse terrains, including urban, mountainous, and forested environments [153]. Zhang et al. [154] introduced a micro-UAV bionic dynamic path planning algorithm that integrates deep neural network optimization/filtering with hawk-eye vision. Their experimental results demonstrated that their UAV could navigate a 20 cm wide corridor and successfully avoid obstacles ranging from 7 to 18 cm. Compared to traditional algorithms such as Multi-scale Directional Patch (MOPs) + Scale-Invariant Feature Transform (SIFT) and the Size Extension Algorithm (SEA), their approach achieved over 20. Akshya et al. [155] improved UAV path planning efficiency using a deep neural network optimized by the Adam algorithm. Their experiments showed that models employing the sigmoid activation function performed well in terms of the mean square error (MSE). When using the ReLU activation function, the errors in the X and Y coordinates were reduced by 36.5% and 24.9%, respectively, while the root mean square error (RMSE) decreased by 20.1% and 17.5%. Despite their strong performance, deep neural networks generally demand substantial training data and computational power. The intensive resource requirements of the training process can hinder real-time application and limit interpretability, posing challenges for deployment in emergency situations or scenarios where high reliability is essential.

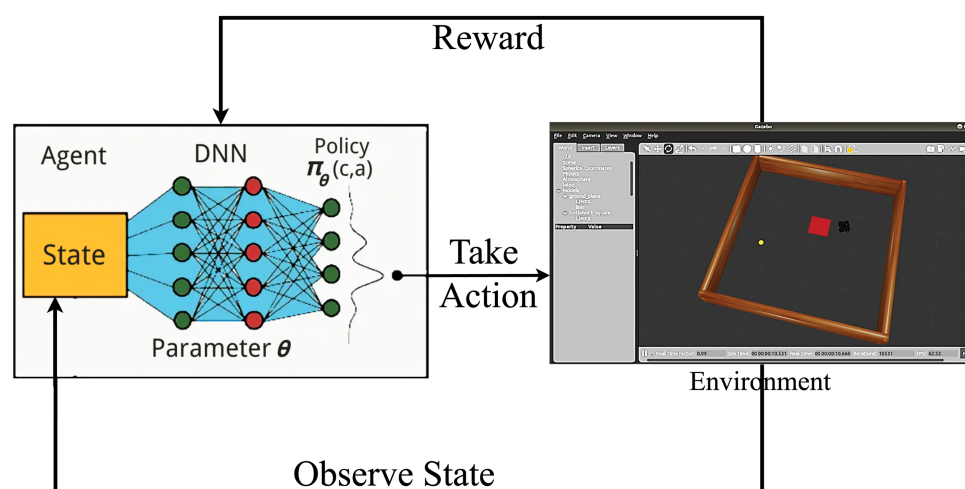


Figure 10. Deep neural networks (DNNs) for UAV path planning task [156].

Physics-Informed Neural Networks (PINNs) [157] represent a novel paradigm in machine learning, integrating physical laws described by partial differential equations directly into the training process of neural networks. In the context of path planning and control, PINNs enable data-driven models to respect underlying physical constraints, such as vehicle dynamics or environmental interactions, thereby improving generalization and reliability. By embedding domain knowledge into the learning architecture, PINNs can efficiently solve forward and inverse problems, facilitate real-time decision making, and reduce the reliance on large labeled datasets. This approach holds significant promise for enhancing the safety, interpretability, and performance of autonomous systems operating in complex, real-world environments.

Through multi-layer nonlinear mapping, DNNs can effectively extract environmental features from high-dimensional perceptual data and generate path decisions. In terms of architectural design, Zhang et al. [154] used the Inception V3 model as the basis and

trained the monocular image classification (obstacle direction perception) through the transfer learning method. In order to improve the classification accuracy, the paper introduces the Improved Bat Algorithm (IBA) to optimize the learning rate, enabling the final classification accuracy to reach 83.18% on the small sample validation set. In another study, Akshya et al. [155] proposed a deep neural network combined with the Adam optimizer for UAV path planning in regional coverage tasks. This method first builds a training set based on simulated LiDAR sensor data and learns the environmental spatial dynamics through the DNN model. The model structure is as follows: Two hidden layers, with 64 neurons in each layer, and the activation function adopts ReLU; there are two units in the output layer; the loss function adopts the mean square error (MSE), and the optimizer is Adam (Adaptive Learning Rate Optimization). In the experiment, the researchers compared the effects of different activation functions (ReLU, tanh, sigmoid) on the model performance. The results showed that the accuracy of trajectory generation was the best when using the sigmoid activation function, and both MSE and RMSE decreased by approximately 20–36.5% compared with ReLU activation. Although DNNs can improve the perception and decision-making accuracy of path planning, there are still many failure risks in practical applications: Environmental change sensitivity: Zhang et al. [154] pointed out that when a DNN encounters dynamic environmental changes during the testing phase (such as the sudden appearance of small obstacles), the success rate of path planning decreases by 10–15%, exposing the model's insufficient adaptability to changes in environmental distribution. In addition, Akshya et al. [155] observed that when using inappropriate activation functions (such as ReLU), the trajectory smoothness decreased significantly, and trajectory jumps or sudden changes occurred. Moreover, if directly applied to dynamic environments (such as sudden obstacles), the current DNN model does not demonstrate good real-time adaptability. These problems suggest that although DNNs perform well in specific controlled environments, in complex, dynamic and highly uncertain actual flight missions, they still need to be improved through domain adaptation, model lightweighting, and rapid perception update mechanisms.

4.3.2. Convolutional Neural Network (CNN)

The strengths of convolutional neural networks (CNNs) in image processing make them a vital technology for UAV path planning, particularly in visual navigation tasks [158,159]. In this context, CNNs can process real-time images captured by UAV cameras, detect obstacles, and optimize flight trajectories, as depicted in Figure 11. For instance, Bae et al. [160] developed a multi-robot path planning algorithm that integrates Deep Q-learning with CNNs. By leveraging CNNs to interpret environmental imagery and reinforcement learning for decision making, their robots demonstrated flexible and efficient movement across various environments. Their experimental results indicated that this approach surpassed traditional algorithms in terms of search time and overall effectiveness. Li et al. [161] presented a CNN-based method for UAV path planning and navigation, combining deep learning to enhance trajectory optimization in real-world scenarios. The CNN component analyzes obstacles in urban landscapes and subsequently generates optimal flight paths [162]. Despite these advancements, CNNs face challenges when dealing with dynamically changing environments and varying lighting conditions, which can impact the accuracy and reliability of path planning across different scenarios. Addressing these issues and enhancing the robustness and real-time capabilities of CNN models remain crucial for further optimizing their application in UAV path planning.

CNNs effectively extract environmental visual features through local receptive fields and convolution mechanisms. For example, the multi-robot path planning method proposed by Bae et al. [160] uses 80×80 grayscale images input by a monocular camera,

extracts features through two convolutional layers (using 32 and 64 3×3 convolutional kernels, respectively, with a stride of 1), and then makes path decisions through two fully connected layers. The convolutional layer adopts the ReLU activation function and introduces the dropout mechanism to suppress overfitting, and combines this with DQN for action selection. Experiments were conducted on the Gazebo simulation platform. The results showed that in a complex obstacle environment, compared with the traditional Q-learning method, the navigation success rate increased by 12%. In terms of hyperparameters, the size of the convolution kernel, step length, pooling strategy, etc., directly affect the quality of feature extraction and the speed of model inference. If the stride is too large, it will lead to the loss of fine-grained obstacle information. If the convolutional layer is too shallow, it may not be sufficient to model complex environmental changes. However, in actual complex environments (such as dynamic occlusion or low-light conditions), the robustness of CNN feature extraction is still limited. The research by McGuire et al. [163] further confirmed that when the illumination changes sharply or the background changes dynamically, the perception accuracy of CNNs decreases by 25–47%, and the failure rate of path planning increases significantly. This indicates that the traditional convolution architecture is highly sensitive to changes in the input distribution and relies on a large amount of diverse training data to enhance the generalization ability of the model.

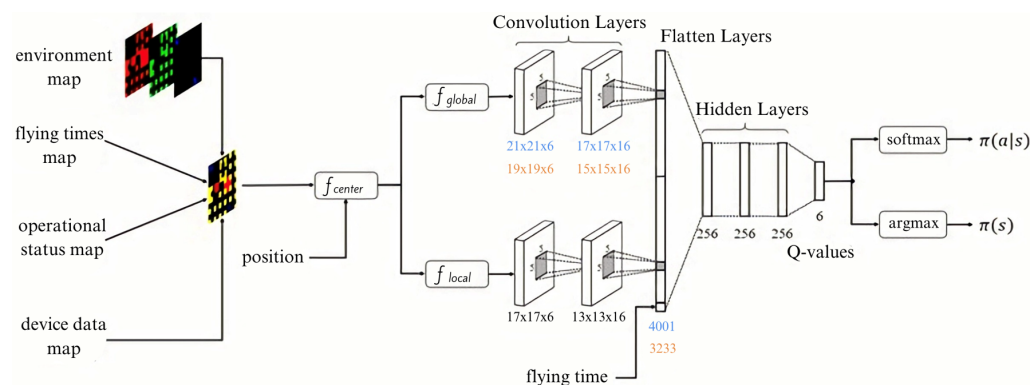


Figure 11. Convolutional neural network (CNN) for UAV path planning task [148].

4.3.3. Generative Adversarial Network (GAN)

Generative adversarial networks (GANs) can produce diverse and natural path planning solutions through adversarial training between generator and discriminator models, as illustrated in Figure 12. By generating multiple flight paths based on mission requirements and selecting the optimal one through the discriminator, GANs are particularly useful in complex urban environments, where they help ensure the safety and effectiveness of UAV routes [164,165]. Eskandari et al. [166] introduced GANs as UAV path planners for real-time navigation in reconfigurable intelligent surface (RIS)-assisted wireless networks. Their approach enables UAVs to optimize flight paths while enhancing communication reliability. Similarly, Gan et al. [167] addressed intelligent UAV (IUAV) path planning in complex environments using a multi-objective projection algorithm. They employed GAN-based environment modeling to simulate realistic navigation scenarios, thereby improving UAV adaptability during flight. However, GANs can be unstable during training, requiring careful balancing between generation and discrimination to maintain training stability and result reliability. Moreover, the use of GANs in path planning remains in the exploratory phase, with ongoing challenges related to improving the quality of generated paths—especially in highly complex scenarios.

Through the adversarial training mechanism of generators and discriminators, GAN can learn the path distribution in complex environments and achieve diverse and efficient

track generation. For instance, Eskandari et al. [166] proposed a GAN-based unmanned aerial vehicle (UAV) path generation method for optimizing flight trajectories in reconfigurable intelligent surface (RIS)-assisted communication systems. This method uses LSTM as the generator and MLP as the discriminator. During the training process, the Adam optimizer (with a learning rate of 10^{-4} and a batch size of 64) is adopted, and it undergoes 5000 rounds of iterative training. The experimental results show that the trajectories generated by the GAN improve the communication reliability index by approximately 15% compared with the traditional heuristic methods, while the time required for path planning is reduced by 30%. Another study, Gan et al. [167] proposed the multi-objective optimization GAN (MO-GAN) method, combining CNN and LSTM structures, to model the path generation problem in complex urban environments. Their experiments show that, compared with the traditional A* algorithm, MO-GAN achieves an excellent performance of shortening the path distance by 18% and increasing the obstacle avoidance success rate by 23% in the simulated urban environment, demonstrating the potential of GANs in handling multi-constraint path planning tasks in complex terrains. Although GANs perform well in the generation of unmanned aerial vehicle paths, there are still problems such as poor training stability and mode collapse. For example, in the experiment of Eskandari et al. [166], if the discriminator is trained too fast, the generator has difficulty learning the effective distribution, resulting in a lack of diversity in the generation path. Furthermore, GANs are highly sensitive to the distribution of training data. If the training set lacks extreme environment samples (such as high-density obstacle areas), the generator often fails to cover all scenarios well.

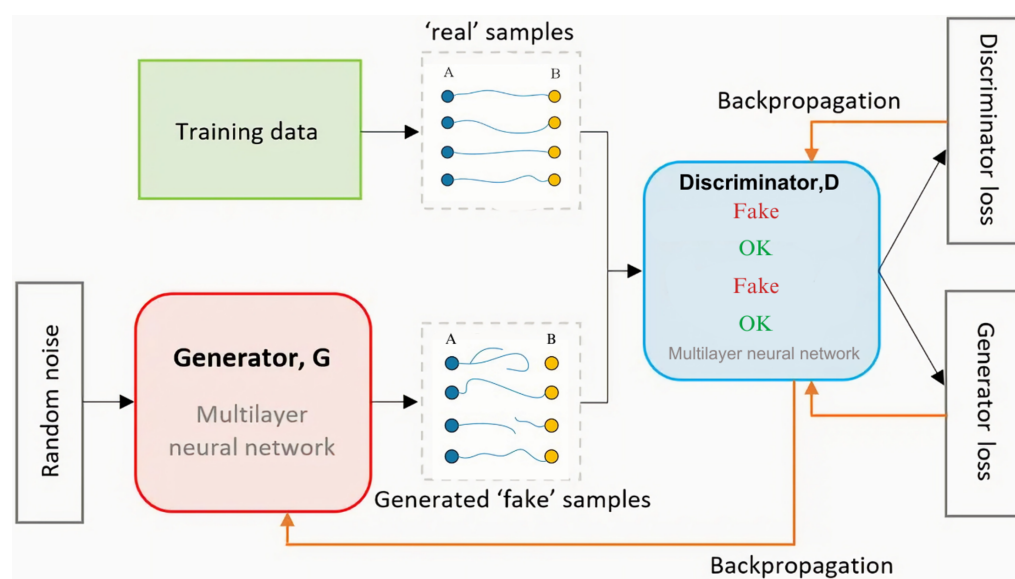


Figure 12. Generative adversarial network (GAN) for UAV path planning task [168].

4.3.4. Critical Analysis

The learning methods show significant advantages over traditional heuristic and optimization algorithms in terms of dynamic environment adaptability, complex scene feature modeling, and autonomous decision-making ability. Table 6 illustrates a comparative analysis of DNN, CNN, and GAN methods for UAV path planning based on key attributes such as network structure, training methods, advantages, disadvantages, and typical application scenarios. Furthermore, the analysis of typical failure cases indicates that in cases such as unoptimized hyperparameters, changes in the distribution of input features (distribution shift), and extreme dynamic changes in the environment, the AI path plan-

ning system may experience performance degradation, convergence difficulties, and even decision-making errors.

Table 6. Comparative analysis of DNN, CNN, and GAN methods.

Attribute	DNN	CNN	GAN
Core Function	High-dimensional feature extraction; trajectory prediction	Image feature extraction; obstacle recognition	Trajectory generation; environment simulation
Network Structure	Multi-layer fully connected network (FC)	Convolution and pooling network (Conv + Pool)	Generator–discriminator adversarial architecture
Training Method	Supervised learning (regression/classification)	Supervised learning (visual navigation)	Unsupervised adversarial training
Advantages	Fast convergence; high trajectory continuity	Accurate perception; strong local feature modeling	Diversity generation; augmenting training data
Disadvantages	Error accumulation; poor adaptability to dynamic environments	Sensitive to lighting/occlusion; inference latency	Unstable training; risk of mode collapse
Suitable Scenarios	Static area coverage; trajectory prediction	UAV visual obstacle avoidance; visual navigation	Complex environment simulation; reinforcement learning assistance

5. Energy-Aware Path Planning

Energy consumption optimization in UAV path planning aims to minimize total energy consumption E_{total} while ensuring efficient and sustainable task execution. This involves optimizing the flight trajectory $(x(t), y(t), z(t))$, taking into account dynamic power consumption, environmental factors, and path rationality. The optimization problem can be expressed as

$$\min E_{\text{total}} = \int_0^T P(v(t), a(t), \text{env}(t)) dt$$

where $P(v(t), a(t), \text{env}(t))$ represents the power consumption as a function of velocity $v(t)$, acceleration $a(t)$, and environmental factors $\text{env}(t)$; and T is the total flight time. The optimization balances energy efficiency, trajectory feasibility, and environmental influences to extend flight time and reduce energy consumption. Through reasonable energy management and path optimization, the flight time can be effectively extended and the energy consumption can be reduced, so as to improve the efficiency and sustainability of the task execution. Energy optimization path planning not only focuses on the shortest path, but also considers the dynamic power consumption of the aircraft, the influence of environmental factors, and the rationality of the flight trajectory.

5.1. Energy Model and Optimization Objective

The energy model of a UAV provides the theoretical foundation for path planning, and the goal of energy optimization is to minimize the total energy required during the flight. Through the construction of reasonable energy model, the energy consumption of UAV during flight can be analyzed in detail, so as to provide data support for path optimization. The energy consumption model of a UAV typically includes two main components: flight power consumption and static power consumption.

Flight power consumption refers to the energy consumed by the UAV during flight in order to overcome factors such as air resistance, gravity and propulsion force. Flight power consumption is not only affected by the efficiency and quality of the power system of the aircraft itself, but also changes with the changes in flight speed, altitude, aerodynamic characteristics, and other factors. In ref. [30], the researchers took an Intel Aero Ready to Fly Drone as the object and a self-made digital multimeter, and conducted experiments in a variety of scenarios to analyze the impact of various factors on UAV energy consumption. The study found that different flight scenarios have different energy consumption charac-

teristics: take-off energy consumption is related to speed; hover, horizontal, and vertical motion energy consumption is stable; and horizontal flight energy consumption is higher. Flight power consumption can be estimated using mathematical models. For fixed-wing UAVs, the Breguet formula estimates energy consumption based on factors like weight, speed, altitude, and fuel efficiency. Yan et al. [169] developed a rotor-wing UAV energy model considering acceleration, deceleration, and time, validated by experiments. Abeywickrama et al. [30] proposed a model based on battery performance, quantifying various factors to predict energy use and optimize mission planning.

Static energy consumption refers to the energy consumed by the UAV to maintain the normal operation of various systems in the non-flight state. This includes the loss of battery standby, energy consumption of sensors and communication systems, etc. The impact of static energy consumption becomes more and more important as the length and complexity of the task increases. For example, in a long-duration airborne mission, static energy consumption may account for a larger percentage of total energy consumption. Therefore, when designing the energy management strategy, we must consider how to reduce the static energy consumption to the greatest extent by adjusting the standby mode and optimizing the working cycle of the sensor.

5.2. Energy Perception Path Planning Algorithm

Energy-sensing path planning algorithms aim to optimize UAV flight trajectories to minimize energy consumption while ensuring task completion. For instance, Na et al. [170] proposed an optimal energy path planning method for UAVs based on an improved particle swarm optimization (PSO) algorithm. By designing a 3D path planning approach and incorporating a parameter-adaptive method based on deep deterministic policy gradient (DDPG), their method addresses the traditional PSO algorithm's tendency to become trapped in local optima when dealing with complex problems. Shivgan et al. [21] formulated UAV path planning as a traveler–dealer problem to further optimize energy consumption. They employed a genetic algorithm to solve the optimization problem, minimizing UAV energy usage primarily by reducing the number of turns in the trajectory. Their experimental results demonstrated that their algorithm significantly lowers energy consumption compared to greedy algorithms, with the advantages becoming more pronounced as the number of waypoints increases. Cabreira et al. [171] introduced an energy-sensing spiral coverage path planning algorithm (E-Spiral) for drone photogrammetry applications. This algorithm takes into account the camera sensor and flight altitude, applying different optimal speeds to reduce energy consumption. Compared to traditional reverse and spiral flight modes, their experimental results show that the E-Spiral algorithm effectively saves energy across various scenarios, particularly in complex terrains and large-scale areas. These algorithms typically maximize energy efficiency through various strategies, as summarized in Table 7.

Table 7. Energy optimization strategies for UAVs.

Aspect	Description
Minimize flight distance	Calculating the shortest path from the starting point to the target point can effectively reduce the total energy consumption of the aircraft [172].
Flight path optimization	Optimizing flight speed to minimize air resistance and improve the efficiency of propulsion systems is a key part of energy optimization [173].
Dynamic adjustment of flight strategy	Real-time missions require UAVs to dynamically adjust their paths based on environmental changes, such as wind variations or obstacles [60].

5.3. Environmental Factors

For UAVs operating outdoors, environmental factors that can significantly affect the energy consumption of UAVs must be considered [174,175], such as wind, air flow,

temperature, etc., among which the influence of wind and air flow is particularly important. Under certain conditions, these factors can effectively reduce or increase the energy consumption of the UAV.

- **Wind**

Wind speed and direction are the key factors affecting UAV flight energy consumption, which directly determines the flight energy consumption. During actual flight, wind speed and direction change, and to deal with these changes, UAVs can sense the wind in real-time through integrated sensors such as anemometers, GPS and IMUs, and adjust their flight path based on the measured wind speed and direction to optimize energy consumption. As mentioned in ref. [176], fixed-wing UAVs utilize wind speed measurement equipment (such as conventional pitot tubes or porous pitot tubes) combined with inertial measurement units (IMUs) to estimate wind speed. In the case of headwinds, the flight path is adjusted to a low-wind-speed area or by adjusting the climb angle to reduce energy consumption, and similar optimization strategies are often implemented with an extended Kalman filter (EKF). In addition, the UAV also analyzes weather data in real time to predict changes in wind speed over a period of time to plan the optimal flight path. For example, a stochastic process model is proposed in ref. [177,178] to describe changes in wind speed, which can help adjust flight strategies according to the predicted wind speed during flight.

- **Air density**

Air density changes with height. Generally, in higher airspace the air is less dense, which means that the vehicle needs to provide more thrust and lift to maintain stable flight, thus consuming more energy. UAV flight path planning needs to take this into account to select the flight altitude with the least energy consumption. For example, in order to solve the path planning problem of electric vertical take-off and landing UAVs (eVTOL UAV) in urban areas, Li et al. [179] proposed a planning method that considers minimum energy consumption. Based on the change in air density with height, a more accurate calculation formula of battery energy consumption is derived and, combined with the improved Voronoi diagram, Dubins geometric curve, and the Floyd algorithm, a safe, shortest, and obstacle-free path is planned. Through simulation analysis, it is found that the path length planned by this method is shorter than that of particle swarm optimization algorithm, which can effectively reduce energy consumption and increase the transportation range of UAVs.

- **Temperature**

Changes in temperature have a significant effect on the performance of aircraft. Temperature not only affects the density of the air, but also directly affects the performance of the drone battery and the efficiency of the engine. In a low-temperature environment, the battery performance of the UAV decreases and the discharge ability weakens, resulting in an increase in energy consumption [180]. Therefore, under different temperature conditions, path planning needs to accurately predict the performance changes of the battery to ensure the stability of the energy supply. Kim et al. [181] proposed a quadratic function model by using the historical and predicted temperature data of the operating area of UAVs and taking into account the effect of UAVs' flight height on temperature by fitting the data with a regression analysis method, and verified that this model was statistically significant and had high reliability.

5.4. Case Study

Energy consumption optimization path planning in actual missions can significantly improve the mission efficiency and economy of UAVs, as shown in Figure 13. Here are some typical application cases.

- **Distribution tasks**

In the field of drone delivery, path planning must not only find the shortest path, but also be as energy efficient as possible [182]. Kevin et al. [180] proposed a multi-travel vehicle routing problem (MTVRP) for UAV delivery, and verified the energy consumption model of a multi-rotor UAV through experiments, and aimed at UAV delivery problems (DDPs). In this paper, the minimum-cost drone delivery problem (MC-DDP) and minimum-time drone delivery problem (MT-DDP) are proposed and the simulated annealing (SA) algorithm is used to solve these problems. The experiments show that SA is close to the optimal solution for both large- and small-scale problems. At the same time, it is found that the cost and delivery time limit, the total delivery time, and the budget are inversely exponential. Reuse of UAVs and optimization of battery weight can significantly reduce the cost and shorten the delivery time.

- **Agricultural monitoring**

In agricultural monitoring missions, drones need to cover large areas of farmland and collect data [183]. By using a path planning algorithm based on energy optimization, the aircraft can reasonably plan a flight path to avoid repeated flights and unnecessary climbs, thus reducing energy consumption and improving operational efficiency. Based on the perception of terrain and wind, path planning can be adjusted in real time to avoid energy-intensive flights in areas with high wind speeds.

- **Disaster relief**

In disaster relief, drones often need to fly for long periods of time and face complex environmental conditions. Through energy optimization path planning, the aircraft can adjust the flight path according to real-time meteorological data, avoiding high winds and unstable airflow areas, thereby reducing energy consumption and improving rescue efficiency.

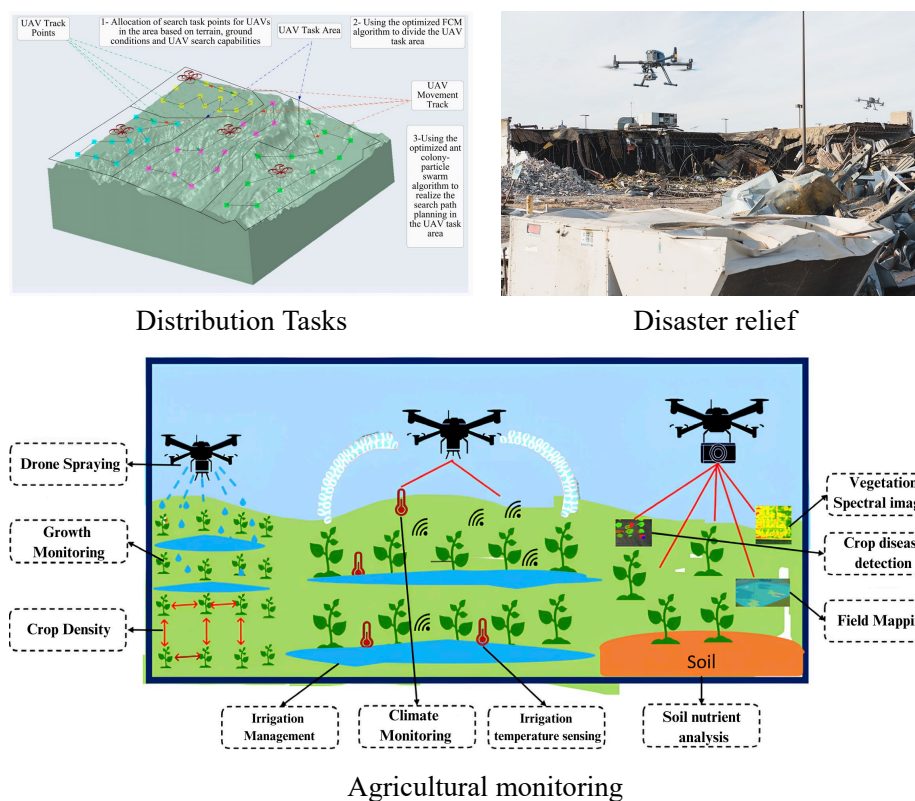


Figure 13. Application cases in energy consumption optimization path planning.

6. Multi-UAV Collaborative Path Planning

Multi-UAV path planning aims to optimize the trajectories of multiple unmanned aerial vehicles (UAVs) while ensuring collision avoidance, minimizing energy consumption, and achieving mission-specific objectives, such as target coverage or surveillance, as shown in Figure 14. The problem is typically formulated as a multi-objective optimization problem, where the cost function combines factors such as distance traveled, energy usage, and task efficiency, subject with constraints like UAV dynamics, communication range, and obstacle avoidance [184]. Mathematically, this can be expressed as

$$\min_{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n} \sum_{i=1}^n (J_{\text{energy}}(\mathbf{P}_i) + J_{\text{time}}(\mathbf{P}_i) + J_{\text{task}}(\mathbf{P}_i))$$

subject to: $\mathbf{P}_i(t) \neq \mathbf{P}_j(t), \forall i \neq j$, $\mathbf{P}_i(t) \notin \mathcal{O}$, and UAV dynamics constraints.

where

- $\mathbf{P}_i$ is the path of UAV i ;
- $J_{\text{energy}}, J_{\text{time}}, J_{\text{task}}$ are the cost terms for the energy, time, and task-specific objectives;
- $\mathcal{O}$ represents obstacles, and the constraints ensure collision avoidance and feasibility.

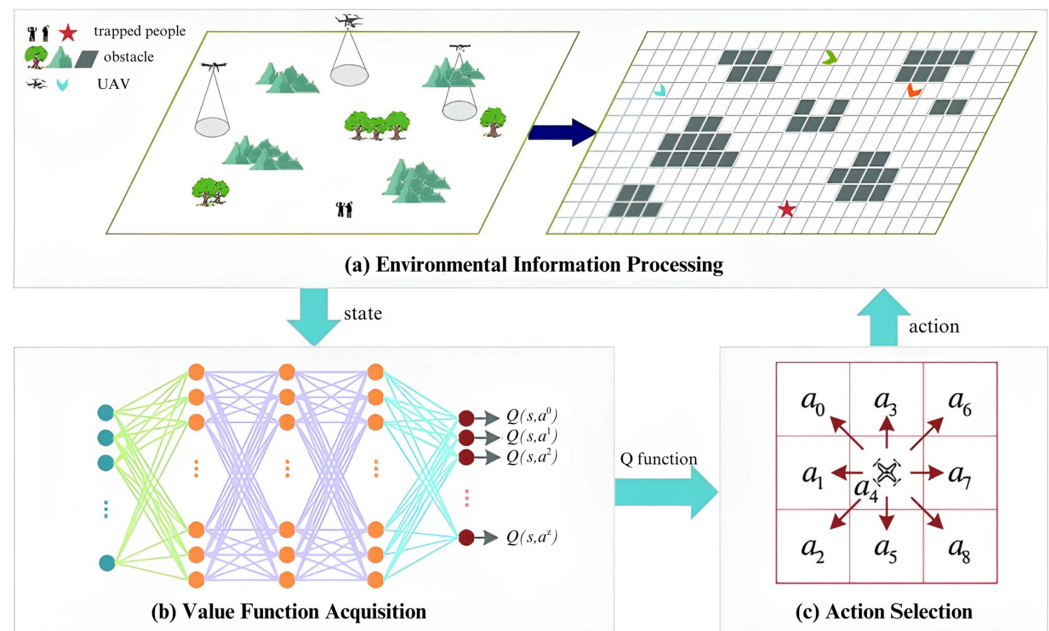


Figure 14. An example of multi-UAV collaborative path planning [185].

6.1. Features of Collaborative Path Planning

Collaborative path planning enables multiple UAVs to work together efficiently by optimizing task allocation, reducing redundant efforts, and ensuring obstacle avoidance. Key features include task allocation, communication constraints, and collaborative obstacle avoidance. This approach enhances mission efficiency by leveraging shared information and coordinated strategies, making it ideal for complex scenarios like disaster relief, agricultural monitoring, and urban security, where teamwork and adaptability are critical for success. The features of collaborative path planning include the following.

- **Task allocation**

In multi-UAV collaboration, task assignment is a key step in path planning, optimizing resources and reducing flight time and energy. Strategies are categorized as optimization-based or demand-based. The Performance Impact (PI) algorithm, a demand-based method, evaluates inclusion (IPI) and removal (RPI) impacts, consid-

ering task cost and priority. Optimization-based methods include exact techniques, heuristics, and hybrids [186]. Task assignment also considers UAV battery levels, sensor capabilities, and task urgency [187].

- **Communication constraints**

Multi-drone collaboration relies on continuous communication for mission updates, flight status, and environmental data. However, dynamic and interference-prone networks pose challenges. Meng et al. [188] address this with the LR-PI algorithm, enhancing task allocation by considering communication delays and packet loss. Additionally, UAV network topology must adapt to changing environments.

- **Collaborative obstacle avoidance**

Traditional path planning focuses on single-UAV obstacle avoidance, while multi-UAV collaborative planning considers dynamic avoidance among multiple aircraft. Rule-based strategies adjust speed and altitude, as seen in Zhang et al. [189], who combined optimal consensus control, particle swarm optimization, and an improved artificial potential field (APF). Path prediction methods anticipate collisions and adjust trajectories. Learning-based approaches, like Luo et al. [190], who used Deep-SARSA, train neural networks for reliable path planning and collision avoidance in dynamic environments.

6.2. Representative Methods

The algorithms for multi-UAV collaborative path planning can be categorized into two main approaches, centralized and distributed, each with its own advantages, limitations, and suitable application scenarios, as shown in Table 8. The centralized framework offers the advantage of generating globally optimal path planning solutions. However, due to its high computational complexity, it becomes challenging to maintain real-time performance, particularly in large-scale applications involving complex tasks. In contrast, the distributed framework enhances system robustness and scalability, making it more resilient to single-node failures and better suited for large-scale applications. However, since each UAV makes decisions based only on local information, achieving a globally optimal solution remains a significant challenge [191].

Table 8. Comparison of centralized and distributed multi-UAV path planning methods.

Aspect	Centralized Path Planning	Distributed Path Planning
Advantages	• Generates globally optimal solutions • Suitable for tasks requiring global optimization • Enables orderly collaboration and system synergy	• Enhances robustness and scalability • Resilient to single-node failures • Reduces computational and communication overhead • Suitable for large-scale, multi-task collaborative scenarios
Limitations	• High computational complexity • Communication delay and bottlenecks • Challenges in real-time performance for large-scale tasks	• Lack of global optimization • Decisions based only on local information
Key Methods	• RRT and its improved versions • PRM and its improved versions • Improved APF • Task scheduling and global path optimization [192,193]	• Market-based mechanisms (e.g., bidding and task allocation) [194] • Metaheuristic approaches [195] • Machine learning techniques (e.g., reinforcement learning) [196]
Application Scenarios	• Disaster relief missions requiring global coordination • Scenarios demanding synchronized task execution	• Large-scale applications with dynamic task requirements • Collaborative multi-task scenarios

6.3. Typical Applications of Multi-UAV Collaboration

The advantage of multi-UAV collaborative path planning is its ability to provide more efficient solutions in multiple domains. The following are some typical application scenarios.

- **Disaster rescue**

In disaster rescue scenarios, multiple UAVs collaborate on tasks like regional search, material delivery, and personnel positioning. Each drone plans its path independently using mission requirements and real-time data, while coordinating to avoid obstacles and improve efficiency. Xiong et al. [197] addressed supply delivery in disaster areas by modeling flight distance, payload, terrain, and threats. They used an adaptive genetic algorithm (AGA) for task assignment and sine-cosine particle swarm optimization (SCPSO) for path planning. Chandran et al. [198] proposed drones as communication relays to establish temporary networks, exploring architecture, routing, and security for multi-UAV disaster monitoring.

- **Area coverage**

In agricultural monitoring, multi-drone collaboration covers large areas efficiently compared to single drones. Ju et al. [31] proposed a distributed group control algorithm for multi-UAV systems, evaluated through metrics like total time, flight time, and battery consumption. The results showed better performance over single UAVs, addressing battery shortages, reducing working time, and improving efficiency. Multi-UAVs can dynamically adjust paths, avoid redundant coverage, and optimize mission execution, offering valuable insights for advancing agricultural technologies.

- **Monitoring patrol**

In border patrol and urban security, multi-drone collaboration enables real-time, continuous surveillance over large areas. Drones perform distinct tasks or alternate patrols while using real-time path adjustments and obstacle avoidance. Muñoz et al. [199] tackled urban multi-UAV path planning with two algorithms: one uses round-trip movements, and the other connects landmark points. Both rely on approximate cell decomposition and the Fast Marching Square (FM^2) strategy for obstacle avoidance, maintaining triangular formations. Simulations showed the algorithms could generate safe, smooth paths and reduce computation time, though challenges remained with large obstacles and real-world validation.

7. Future Directions

As an important part of UAV autonomous navigation, path planning technology is facing new challenges and opportunities with the development of UAV application field and technological progress. For the future, I think the following three parts are the main directions.

7.1. Advanced Optimization of Algorithm Performance

The future development of path planning algorithms needs to find a reasonable balance between computational efficiency and robustness, which urgently needs complex UAV applications. With the wide application of UAVs in various fields, their working environment is becoming more complex and dynamic, and the actual working situation constantly tests the ability of traditional path planning algorithms in computational and real-time environments [200]. Therefore, how to optimize the computational performance of the algorithms under the premise of path planning quality and improve their adaptability in complex environments will become the focus of future research. The following two points must be considered.

- **Enhancement of computing efficiency**

Optimize the computing speed and resource consumption of existing path planning algorithms, especially in large-scale tasks and complex environments, to ensure that path planning can be executed in real time. In the tasks of large-scale cluster control, multi-objective path planning, and high-dimensional-space search, the traditional algorithms such as the classical graph search algorithm have difficulties in meeting the real-time requirements because of the large amount of computation. According to the advantages of each algorithm's principles and the complementarities among algorithms, hybrid algorithms have shown great potential in UAV path planning in recent years. For example, Liu et al. [201] proposed an improved artificial potential field (APF) UAV path planning algorithm (G-APF) guided by Rapidly Exploring Random Trees (RRT) based on environment perception modeling. By combining the advantages of the RRT and APF algorithms and introducing environment awareness, the advantages of each algorithm are utilized to overcome the limitations of traditional methods and optimize the computational efficiency of the existing path planning algorithm.

- **Improvement of robustness**

In UAV applications, path planning algorithms need to deal with dynamic and complex environments, such as sudden obstacles, communication delays, target location changes, and other uncertainties that will have an impact on the planning results. The existing path algorithms often lack sufficient foresight when dealing with emergencies, which can easily lead to planning failure, or they need a short time to recalculate the path. Therefore, one of the important research directions is to improve the robustness of the algorithms so that they can maintain stable programming under dynamic environments. To solve this problem, some effective methods have been introduced so that path planning can be adjusted in real-time according to environmental changes. For example, Zhang et al. [202] proposed an improved adaptive gray wolf optimization (AGWO) algorithm to solve premature convergence and local optima issues in GWO. By designing an adaptive weight factor based on centrifugal distance change, the algorithm dynamically updates positions and improves robustness. Ramezani et al. [203] introduced the SERT-DQN+ algorithm, which ensures system stability by addressing UAV failures and communication interruptions in multi-UAV systems.

7.2. Edge Computing, Swarm Intelligence, and Safety

Edge computing and onboard inference are rapidly becoming indispensable for unmanned aerial vehicles (UAVs), especially as these platforms are deployed in increasingly complex and dynamic environments [204–206]. Traditionally, UAVs relied on transmitting raw data to centralized cloud servers for processing, which introduces latency and bandwidth limitations. With edge computing, however, data processing and inference can occur directly on the UAV itself, enabling real-time decision making and reducing dependence on unreliable or delayed communication links. This is particularly crucial in scenarios such as disaster response or military reconnaissance, where immediate action based on sensory input is required. Onboard inference, powered by lightweight AI models optimized for embedded hardware, allows UAVs to detect obstacles, identify objects, and adapt their flight paths autonomously, thereby enhancing their operational effectiveness and resilience.

The integration of swarm intelligence with federated learning [207,208] further amplifies the capabilities of UAV fleets. Swarm intelligence enables multiple UAVs to coordinate their actions, share information, and collectively solve tasks that would be infeasible for a single drone. When combined with federated learning, each UAV can locally train its AI models using its own data and periodically share only the model updates with other drones or a central aggregator. This approach not only preserves data privacy and reduces

communication overhead but also allows the entire swarm to rapidly adapt to new environments or mission objectives. The synergy between swarm intelligence and federated learning fosters a robust, scalable, and adaptive system, where knowledge is distributed and continuously refined without exposing sensitive raw data.

However, as UAVs become more autonomous and intelligent, ensuring safety and adhering to ethical constraints in navigation is paramount [209]. Autonomous navigation systems must be designed to avoid not only physical hazards such as collisions with obstacles and other aircraft but also to respect no-fly zones, privacy boundaries, and local regulations. Ethical considerations extend beyond technical safety to include issues such as surveillance, data protection, and the potential for misuse in sensitive areas. Implementing transparent decision-making processes, fail-safe mechanisms, and robust validation protocols is essential to build public trust and ensure responsible deployment. As UAV technology evolves, it is critical that engineers and policymakers collaborate to establish comprehensive frameworks that balance innovation with societal values and legal requirements, ensuring that autonomous UAV operations remain safe, ethical, and beneficial to all stakeholders.

7.3. Integration of Deep Learning and Path Planning

In recent years, deep learning technology has achieved revolutionary breakthroughs in fields such as computer vision and speech recognition, and has gradually expanded into areas including robotics, autonomous systems, and intelligent control [210,211]. Looking ahead, deep learning is expected to play a pivotal role in the field of path planning. The core advantage of deep learning lies in its powerful data-driven capabilities, enabling it to extract latent patterns from vast amounts of environmental data and make accurate predictions and decisions. Unlike traditional path planning methods, which often depend on predefined models or manually crafted rules, deep learning approaches can train deep neural networks (DNNs) to allow autonomous systems to optimize paths directly from sensory data. The potential of data-driven methods is especially significant in dynamic environments, as neural networks can process information from multiple sensors in real time and continuously learn and adapt path planning strategies. However, current algorithms still face challenges, such as overestimation or underestimation of Q-values, prolonged exploration periods, and sparse rewards. Addressing and optimizing these issues represents a key direction for future research. For example, Luo et al. [212] proposed an improved algorithm, I-TD3, based on the average TD3 algorithm and prioritized experience replay, to overcome the limitations of traditional UAV path planning in complex, dynamic environments. Their approach modifies the TD3 experience buffer to use prioritized experience replay, introduces an average TD3 algorithm to avoid Q-value underestimation, and designs a new reward function to tackle the problem of sparse rewards.

7.4. Smart Sensors Fused with Path Planning

The rapid advancement of intelligent sensor technology has provided more abundant and accurate real-time data for path planning, enabling algorithms to adapt more effectively to environmental changes and make dynamic adjustments. Moving forward, autonomous systems will increasingly rely on real-time data to perceive dynamic environments and adjust their paths, resulting in more accurate and robust path planning. As sensor technology continues to evolve, both the variety and precision of sensors are being optimized, allowing for more comprehensive and precise environmental perception. Autonomous systems, such as drones, are typically equipped with various sensors, including LiDAR, radar, and vision sensors (cameras). The real-time data collected by these sensors can be directly integrated into path planning algorithms, enabling drones or robots to rapidly

detect environmental changes and dynamically adjust their trajectories. For example, Abdalmanan et al. [213] proposed the VFH-QL algorithm, which combines Q-learning with a vector field histogram. By equipping the experimental robot with 2D LiDAR and utilizing the sensor data to enhance state-space representation and reward functions, their approach effectively addresses multi-objective path planning in unknown environments. Similarly, Zhang et al. [202] developed a path planning method for UAVs equipped with video cameras for ground area monitoring, achieving effective coverage of target areas even in complex environments. Furthermore, due to the inherent limitations of individual sensors, future research in path planning is expected to focus on multi-sensor fusion technologies to obtain more comprehensive and reliable environmental information [214]. For instance, while camera sensors can provide rich scene information, they may under-perform in low-light or high-glare conditions. LiDAR, on the other hand, offers accurate depth information. The integration of these sensors can enhance environmental perception and improve the reliability of path planning.

7.5. Path Planning Requirements in Special Scenarios

The path planning requirements of UAVs become more complex and challenging in special scenarios. These scenarios often involve extreme environmental conditions or special operational constraints that require greater adaptability and flexibility from path planning techniques. Two common scenarios are mentioned below.

- **Extreme weather conditions:** In bad weather (such as heavy rain, strong wind, haze, etc.), the flight stability and sensor performance of drones will be greatly affected, and traditional path planning algorithms have difficulties in effectively coping [215]. This requires that the algorithm can obtain real-time meteorological data (such as wind speed, air pressure, humidity, etc.) [216], so that, combined with the dynamic characteristics of the aircraft and the energy consumption model, the UAV can optimize the path in bad weather and choose a more stable flight path with lower energy consumption.
- **Navigation without GPS:** In environments where GPS signals are weak or missing (e.g., indoors, underground, urban canyons, etc.), traditional GPS-based path planning methods will fail [217]. This requires the path planning algorithm to be able to integrate other positioning techniques, such as inertial navigation and visual SLAM. In a GPS-free environment, inertial navigation systems (INSs) and visual SLAM (Simultaneous Localization and Mapping) will play an important role. By combining these technologies, the path planning algorithm can make accurate positioning and obstacle avoidance decisions based on real-time sensor information, ensuring that the UAV can still complete its mission in a GPS-free environment. For example, Radwan et al. [218] proposed an end-to-end application that integrates a marker-based visual synchronous positioning and mapping (VSLAM) framework into drones to reconstruct maps and generate 3D scene maps in indoor environments, where GPS signals are missing.
- **High-speed flight:** Recent advances in UAV trajectory planning have also focused on scenarios involving high-speed flight, where traditional path planning and control algorithms may not be sufficient to ensure accurate and robust trajectory tracking. For instance, neural network-based PID controllers [219] have been proposed to enhance the adaptability and performance of UAVs in complex environments by learning optimal control strategies from data, thus improving trajectory tracking accuracy even under dynamic and uncertain conditions. Additionally, comparative studies of optimization algorithms, such as genetic algorithms and stochastic hill climbing, have demonstrated their effectiveness in multi-UAV path planning for inspection missions, particularly when UAVs operate at speeds approaching or exceeding the speed of

sound [220]. These approaches highlight the importance of integrating advanced control and optimization techniques to address the unique challenges posed by special scenarios, ensuring safe, efficient, and reliable UAV operations.

- **UAV-UGV cooperation:** Recent developments in path planning have increasingly emphasized the significance of UAV–UGV cooperation, as such multi-agent systems present unique challenges and opportunities compared to single-platform scenarios. For example, effective path planning in cooperative UAV–UGV systems must address issues such as localization uncertainty, heterogeneous vehicle dynamics, and the need for real-time information sharing in complex or unknown environments. Petrillo et al. [221] explored search planning for UAV/UGV teams operating under localization uncertainty, demonstrating the necessity of robust algorithms for subterranean missions. Similarly, Li et al. [222] proposed a hybrid path planning method that leverages the complementary capabilities of UAVs and UGVs to enhance mission efficiency and adaptability. Recent work has also investigated the use of UAVs for advance reconnaissance to inform UGV path planning in unknown environments [223], as well as multi-agent frameworks for automatic sensory data collection in cluttered settings [224]. These studies highlight the importance of integrated, cooperative planning strategies that account for the distinct constraints and advantages of both aerial and ground vehicles, thereby broadening the scope of path planning research and offering more comprehensive solutions for real-world applications.

8. Conclusions

Path planning technology is a core component of UAV autonomous flight. As UAV applications diversify and operational environments become more complex, advancements in path planning will directly influence the performance and application scope of UAV systems. This paper reviews the main methods and techniques of path planning, analyzing their adaptability in both static and dynamic environments. Furthermore, emerging approaches that integrate machine learning and deep reinforcement learning (such as Q-learning, DQN, and related algorithms) are discussed. These innovative techniques have shown significant promise in enhancing real-time adaptability and managing uncertainty in complex scenarios. Additionally, energy-aware path planning and multi-UAV collaborative strategies are examined, offering theoretical foundations and practical solutions for extending flight endurance, minimizing energy consumption, and improving overall mission efficiency. Looking ahead, with continued optimization of algorithm performance, the integration of deep learning and sensor technologies, and in-depth research into the requirements of specialized scenarios, UAV path planning is expected to achieve significant advancements in efficiency, intelligence, and adaptability. As technology continues to evolve and application domains expand, UAVs will play increasingly important roles in fields such as urban management, disaster emergency response, and logistics and transportation. The ongoing development of path planning technology will propel the UAV field toward higher levels of automation and intelligence, providing crucial support for the future of intelligent transportation, autonomous driving, and related industries.

Author Contributions: All authors have contributed substantially to and agree with the manuscript's content. Conception/design, provision of study materials, and the collection and assembly of data: X.Z., H.G. and L.Z.; data analysis and interpretation: W.M. and X.H.; manuscript preparation: X.Z., X.H. and W.M.; final approval of the manuscript: W.M., X.Z. and L.Z. The guarantor of the paper takes responsibility for the integrity of the work as a whole, from its inception to publication. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported in part by the Natural Science Foundation of Shandong Province (ZR2023QF122), in part by the National Natural Science Foundation of China (62302124), in part by the National Key Research and Development Program of China (2023YFB3307504) and in part by the Youth Teacher Development Foundation of Harbin Institute of Technology (IDGA10002143).

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Santoso, F.; Garratt, M.A.; Anavatti, S.G. State-of-the-art intelligent flight control systems in unmanned aerial vehicles. *IEEE Trans. Autom. Sci. Eng.* **2017**, *15*, 613–627. [\[CrossRef\]](#)
2. Bae, J.; Sohn, K.Y.; Lee, H.; Lee, H.; Lee, H. Structure of UAV-based Emergency Mobile Communication Infrastructure. In Proceedings of the 2021 International Conference on Information and Communication Technology Convergence (ICTC), Jeju Island, Republic of Korea, 20–22 October 2021; pp. 634–636.
3. Daniel, K.; Rohde, S.; Wietfeld, C. Leveraging public wireless communication infrastructures for UAV-based sensor networks. In Proceedings of the 2010 IEEE International Conference on Technologies for Homeland Security (HST), Waltham, MA, USA, 8–10 November 2010; pp. 179–184.
4. Asadzadeh, S.; de Oliveira, W.J.; de Souza Filho, C.R. UAV-based remote sensing for the petroleum industry and environmental monitoring: State-of-the-art and perspectives. *J. Pet. Sci. Eng.* **2022**, *208*, 109633. [\[CrossRef\]](#)
5. Aasen, H.; Honkavaara, E.; Lucieer, A.; Zarco-Tejada, P.J. Quantitative remote sensing at ultra-high resolution with UAV spectroscopy: A review of sensor technology, measurement procedures, and data correction workflows. *Remote Sens.* **2018**, *10*, 1091. [\[CrossRef\]](#)
6. Abubakar, A.I.; Ahmad, I.; Omeke, K.G.; Ozturk, M.; Ozturk, C.; Abdel-Salam, A.M.; Mollel, M.S.; Abbasi, Q.H.; Hussain, S.; Imran, M.A. A survey on energy optimization techniques in UAV-based cellular networks: From conventional to machine learning approaches. *Drones* **2023**, *7*, 214. [\[CrossRef\]](#)
7. Erenoglu, R.C.; Erenoglu, O.; Arslan, N. Accuracy assessment of low cost UAV based city modelling for urban planning. *Teh. Vjesn.* **2018**, *25*, 1708–1714.
8. Islam, N.; Rashid, M.M.; Pasandideh, F.; Ray, B.; Moore, S.; Kadel, R. A review of applications and communication technologies for internet of things (Iot) and unmanned aerial vehicle (uav) based sustainable smart farming. *Sustainability* **2021**, *13*, 1821. [\[CrossRef\]](#)
9. Lottes, P.; Khanna, R.; Pfeifer, J.; Siegwart, R.; Stachniss, C. UAV-based crop and weed classification for smart farming. In Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA), Singapore, 29 May–3 June 2017; pp. 3024–3031.
10. Khan, A.; Gupta, S.; Gupta, S.K. Emerging UAV technology for disaster detection, mitigation, response, and preparedness. *J. Field Robot.* **2022**, *39*, 905–955. [\[CrossRef\]](#)
11. Erdelj, M.; Natalizio, E.; Chowdhury, K.R.; Akyildiz, I.F. Help from the sky: Leveraging UAVs for disaster management. *IEEE Pervasive Comput.* **2017**, *16*, 24–32. [\[CrossRef\]](#)
12. Chaturvedi, S.K.; Sekhar, R.; Banerjee, S.; Kamal, H. Comparative review study of military and civilian unmanned aerial vehicles (UAVs). *INCAS Bull.* **2019**, *11*, 181–182. [\[CrossRef\]](#)
13. Yin, S.; He, R.; Li, J.; Chen, L.; Zhang, S. Research on the operational mode of manned/unmanned collaboratively detecting drone swarm. In Proceedings of the 2021 IEEE International Conference on Unmanned Systems (ICUS), Beijing, China, 15–17 October 2021; pp. 560–564.
14. Song, B.D.; Park, K.; Kim, J. Persistent UAV delivery logistics: MILP formulation and efficient heuristic. *Comput. Ind. Eng.* **2018**, *120*, 418–428. [\[CrossRef\]](#)
15. Zhang, H.-Y.; Lin, W.-M.; Chen, A.-X. Path planning for the mobile robot: A review. *Symmetry* **2018**, *10*, 450. [\[CrossRef\]](#)
16. Sanchez-Ibanez, J.R.; Pérez-del Pulgar, C.J.; García-Cerezo, A. Path planning for autonomous mobile robots: A review. *Sensors* **2021**, *21*, 7898. [\[CrossRef\]](#) [\[PubMed\]](#)
17. Karur, K.; Sharma, N.; Dharmatti, C.; Siegel, J.E. A survey of path planning algorithms for mobile robots. *Vehicles* **2021**, *3*, 448–468. [\[CrossRef\]](#)
18. Huang, G.; Hu, M.; Yang, X.; Wang, X.; Wang, Y.; Huang, F. A Review of Constrained Multi-Objective Evolutionary Algorithm-Based Unmanned Aerial Vehicle Mission Planning: Key Techniques and Challenges. *Drones* **2024**, *8*, 316. [\[CrossRef\]](#)

19. Monwar, M.; Semiari, O.; Saad, W. Optimized path planning for inspection by unmanned aerial vehicles swarm with energy constraints. In Proceedings of the 2018 IEEE Global Communications Conference (GLOBECOM), Abu Dhabi, United Arab Emirates, 9–13 December 2018; pp. 1–6.
20. Yin, C.; Xiao, Z.; Cao, X.; Xi, X.; Yang, P.; Wu, D. Offline and online search: UAV multiobjective path planning under dynamic urban environment. *IEEE Internet Things J.* **2017**, *5*, 546–558. [\[CrossRef\]](#)
21. Shivgan, R.; Dong, Z. Energy-efficient drone coverage path planning using genetic algorithm. In Proceedings of the 2020 IEEE 21st International Conference on High Performance Switching and Routing (HPSR), Newark, NJ, USA, 11–14 May 2020; pp. 1–6.
22. Jin, H.; Jin, X.; Zhou, Y.; Guo, P.; Ren, J.; Yao, J.; Zhang, S. A survey of energy efficient methods for UAV communication. *Veh. Commun.* **2023**, *41*, 100594. [\[CrossRef\]](#)
23. Ruz, J.J.; Pajares, G.; de la Cruz, J.M.; Arevalo, O. *UAV Trajectory Planning for Static and Dynamic Environments*; Citeseer: Princeton, NJ, USA, 2009.
24. Budiyo, A.; Cahyadi, A.; Adji, T.B.; Wahyunggoro, O. UAV obstacle avoidance using potential field under dynamic environment. In Proceedings of the 2015 International Conference on Control, Electronics, Renewable Energy and Communications (ICCEREC), Bandung, Indonesia, 27–29 August 2015; pp. 187–192.
25. De Filippis, L.; Guglieri, G.; Quagliotti, F. Path planning strategies for UAVS in 3D environments. *J. Intell. Robot. Syst.* **2012**, *65*, 247–264. [\[CrossRef\]](#)
26. Dai, R.; Fotedar, S.; Radmanesh, M.; Kumar, M. Quality-aware UAV coverage and path planning in geometrically complex environments. *Ad Hoc Netw.* **2018**, *73*, 95–105. [\[CrossRef\]](#)
27. Kim, J.; Crassidis, J.L. UAV path planning for maximum visibility of ground targets in an urban area. In Proceedings of the 2010 13th International Conference on Information Fusion, Edinburgh, UK, 26–29 July 2010; pp. 1–7.
28. Mesquita, R.; Gaspar, P.D. A novel path planning optimization algorithm based on particle swarm optimization for UAVs for bird monitoring and repelling. *Processes* **2021**, *10*, 62. [\[CrossRef\]](#)
29. Boukoberine, M.N.; Zhou, Z.; Benbouzid, M. A critical review on unmanned aerial vehicles power supply and energy management: Solutions, strategies, and prospects. *Appl. Energy* **2019**, *255*, 113823. [\[CrossRef\]](#)
30. Abeywickrama, H.V.; Jayawickrama, B.A.; He, Y.; Dutkiewicz, E. Comprehensive energy consumption model for unmanned aerial vehicles, based on empirical studies of battery performance. *IEEE Access* **2018**, *6*, 58383–58394. [\[CrossRef\]](#)
31. Ju, C.; Son, H.I. Multiple UAV systems for agricultural applications: Control, implementation, and evaluation. *Electronics* **2018**, *7*, 162. [\[CrossRef\]](#)
32. Skorobogatov, G.; Barrado, C.; Salami, E. Multiple UAV systems: A survey. *Unmanned Syst.* **2020**, *8*, 149–169. [\[CrossRef\]](#)
33. Bellingham, J.; Tillerson, M.; Richards, A.; How, J.P. Multi-task allocation and path planning for cooperating UAVs. In *Cooperative Control: Models, Applications and Algorithms*; Springer: Boston, MA, USA, 2003; pp. 23–41.
34. Moon, S.; Oh, E.; Shim, D.H. An integral framework of task assignment and path planning for multiple unmanned aerial vehicles in dynamic environments. *J. Intell. Robot. Syst.* **2013**, *70*, 303–313. [\[CrossRef\]](#)
35. Qin, P.; Li, J.; Zhang, J.; Fu, Y. Joint Task Allocation and Trajectory Optimization for Multi-UAV Collaborative Air-Ground Edge Computing. *IEEE Trans. Netw. Sci. Eng.* **2024**, *11*, 6231–6243. [\[CrossRef\]](#)
36. Liu, D.; Bao, W.; Zhu, X.; Fei, B.; Men, T.; Xiao, Z. Cooperative path optimization for multiple UAVs surveillance in uncertain environment. *IEEE Internet Things J.* **2021**, *9*, 10676–10692. [\[CrossRef\]](#)
37. López, B.; Muñoz, J.; Quevedo, F.; Monje, C.A.; Garrido, S.; Moreno, L.E. Path planning and collision risk management strategy for multi-UAV systems in 3D environments. *Sensors* **2021**, *21*, 4414. [\[CrossRef\]](#)
38. Yao, P.; Wang, H.; Su, Z. Real-time path planning of unmanned aerial vehicle for target tracking and obstacle avoidance in complex dynamic environment. *Aerosp. Sci. Technol.* **2015**, *47*, 269–279. [\[CrossRef\]](#)
39. Xu, C.; Liao, X.; Tan, J.; Ye, H.; Lu, H. Recent research progress of unmanned aerial vehicle regulation policies and technologies in urban low altitude. *IEEE Access* **2020**, *8*, 74175–74194. [\[CrossRef\]](#)
40. Telli, K.; Kraa, O.; Himeur, Y.; Ouamane, A.; Boumehraz, M.; Atalla, S.; Mansoor, W. A comprehensive review of recent research trends on unmanned aerial vehicles (uavs). *Systems* **2023**, *11*, 400. [\[CrossRef\]](#)
41. MahmoudZadeh, S.; Yazdani, A.; Kalantari, Y.; Ciftler, B.; Aidarus, F.; Al Kadri, M.O. Holistic review of UAV-centric situational awareness: Applications, limitations, and algorithmic challenges. *Robotics* **2024**, *13*, 117. [\[CrossRef\]](#)
42. Xie, R.; Meng, Z.; Wang, L.; Li, H.; Wang, K.; Wu, Z. Unmanned aerial vehicle path planning algorithm based on deep reinforcement learning in large-scale and dynamic environments. *IEEE Access* **2021**, *9*, 24884–24900. [\[CrossRef\]](#)
43. Hein, D.; Kraft, T.; Brauchle, J.; Berger, R. Integrated uav-based real-time mapping for security applications. *ISPRS Int. J. Geo-Inf.* **2019**, *8*, 219. [\[CrossRef\]](#)
44. Schøler, F.; la Cour-Harbo, A.; Bisgaard, M. Generating approximative minimum length paths in 3D for UAVs. In Proceedings of the 2012 IEEE Intelligent Vehicles Symposium, Madrid, Spain, 3–7 June 2012; pp. 229–233.
45. Techy, L.; Woolsey, C.A. Minimum-time path planning for unmanned aerial vehicles in steady uniform winds. *J. Guid. Control Dyn.* **2009**, *32*, 1736–1746. [\[CrossRef\]](#)

46. Elmokadem, T.; Savkin, A.V. A hybrid approach for autonomous collision-free UAV navigation in 3D partially unknown dynamic environments. *Drones* **2021**, *5*, 57. [\[CrossRef\]](#)
47. Chen, J.Y. UAV-guided navigation for ground robot tele-operation in a military reconnaissance environment. *Ergonomics* **2010**, *53*, 940–950. [\[CrossRef\]](#)
48. Alotaibi, E.T.; Alqefari, S.S.; Koubaa, A. Lsar: Multi-uav collaboration for search and rescue missions. *IEEE Access* **2019**, *7*, 55817–55832. [\[CrossRef\]](#)
49. Sun, R.; Zhang, W.; Zheng, J.; Ochieng, W.Y. GNSS/INS integration with integrity monitoring for UAV no-fly zone management. *Remote Sens.* **2020**, *12*, 524. [\[CrossRef\]](#)
50. Jiménez López, J.; Mulero-Pázmány, M. Drones for conservation in protected areas: Present and future. *Drones* **2019**, *3*, 10. [\[CrossRef\]](#)
51. Zhu, L.; Cheng, X.; Yuan, F.G. A 3D collision avoidance strategy for UAV with physical constraints. *Measurement* **2016**, *77*, 40–49. [\[CrossRef\]](#)
52. Pastor, E.; Lopez, J.; Royo, P. UAV payload and mission control hardware/software architecture. *IEEE Aerosp. Electron. Syst. Mag.* **2007**, *22*, 3–8. [\[CrossRef\]](#)
53. Yang, S. Analysis and Optimization of UAV Turning Methodology. 2022. Available online: <https://escholarship.mcgill.ca/concern/papers/k35699091> (accessed on 4 April 2022).
54. Kurdel, P.; Sedláčková, A.N.; Labun, J. UAV flight safety close to the mountain massif. *Transp. Res. Procedia* **2019**, *43*, 319–327. [\[CrossRef\]](#)
55. Sujit, P.; Beard, R. Multiple UAV path planning using anytime algorithms. In Proceedings of the 2009 American Control Conference, St. Louis, MO, USA, 10–12 June 2009; pp. 2978–2983.
56. Zeng, Y.; Zhang, R. Energy-efficient UAV communication with trajectory optimization. *IEEE Trans. Wirel. Commun.* **2017**, *16*, 3747–3760. [\[CrossRef\]](#)
57. Paredes, J.A.; Saito, C.; Abarca, M.; Cuellar, F. Study of effects of high-altitude environments on multicopter and fixed-wing UAVs' energy consumption and flight time. In Proceedings of the 2017 13th IEEE Conference on Automation Science and Engineering (CASE), Xi'an, China, 20–23 August 2017; pp. 1645–1650.
58. Babu, N.; Ntougias, K.; Papadias, C.B.; Popovski, P. Energy efficient altitude optimization of an aerial access point. In Proceedings of the 2020 IEEE 31st Annual International Symposium on Personal, Indoor and Mobile Radio Communications, London, UK, 31 August–3 September 2020; pp. 1–7.
59. Okulski, M.; Ławryńczuk, M. How much energy do we need to fly with greater agility? Energy consumption and performance of an attitude stabilization controller in a quadcopter drone: A modified MPC vs. PID. *Energies* **2022**, *15*, 1380. [\[CrossRef\]](#)
60. Al-Sabban, W.H.; Gonzalez, L.F.; Smith, R.N. Wind-energy based path planning for unmanned aerial vehicles using markov decision processes. In Proceedings of the 2013 IEEE International Conference on Robotics and Automation, Karlsruhe, Germany; 6–10 May 2013; pp. 784–789.
61. Javaid, A. Understanding Dijkstra's Algorithm. 2013. Available online: <https://ssrn.com/abstract=2340905> (accessed on 17 October 2013).
62. Duchoň, F.; Babinec, A.; Kajan, M.; Beňo, P.; Florek, M.; Fico, T.; Jurišica, L. Path planning with modified a star algorithm for a mobile robot. *Procedia Eng.* **2014**, *96*, 59–69. [\[CrossRef\]](#)
63. Karaman, S.; Frazzoli, E. Sampling-based algorithms for optimal motion planning. *Int. J. Robot. Res.* **2011**, *30*, 846–894. [\[CrossRef\]](#)
64. Zammit, C.; Van Kampen, E.J. Comparison of a* and rrt in real-time 3d path planning of uavs. In Proceedings of the American Institute of Aeronautics and Astronautics Scitech 2020 Forum (AIAA), Orlando, FL, USA, 6–10 January 2020; p. 0861.
65. Noreen, I.; Khan, A.; Habib, Z. Optimal path planning using RRT* based approaches: A survey and future directions. *Int. J. Adv. Comput. Sci. Appl.* **2016**, *7*, 97–107. [\[CrossRef\]](#)
66. Kuffner, J.J.; LaValle, S.M. RRT-connect: An efficient approach to single-query path planning. In Proceedings of the 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and Automation. Symposia Proceedings (Cat. No. 00CH37065), San Francisco, CA, USA, 5–8 January 2000; Volume 2, pp. 995–1001.
67. Haupt, R.L.; Werner, D.H. *Genetic Algorithms in Electromagnetics*; John Wiley & Sons: Hoboken, NJ, USA, 2007.
68. Eberhart, R.; Kennedy, J. A new optimizer using particle swarm theory. In Proceedings of the MHS'95—Sixth International Symposium on Micro Machine and Human Science, Nagoya, Japan, 4–6 October 1995; pp. 39–43.
69. Patle, B.; Babu, L. G.; Pandey, A.; Parhi, D.; Jagadeesh, A. A review: On path planning strategies for navigation of mobile robot. *Def. Technol.* **2019**, *15*, 582–606. [\[CrossRef\]](#)
70. Dorigo, M.; Maniezzo, V.; Colnori, A. Distributed optimization by ant colonies. In Proceedings of the European Conference on Artificial Life, Paris, France, 7–10 October 1991; Elsevier Publishing: Paris, France, 1991; Volume 134, pp. 134–142.
71. Abhishek, B.; Ranjit, S.; Shankar, T.; Eappen, G.; Sivasankar, P.; Rajesh, A. Hybrid PSO-HSA and PSO-GA algorithm for 3D path planning in autonomous UAVs. *SN Appl. Sci.* **2020**, *2*, 1805. [\[CrossRef\]](#)

72. Al-Ansarry, S.; Al-Darraj, S. Hybrid RRT-A*: An Improved Path Planning Method for an Autonomous Mobile Robots. *Iraqi J. Electr. Electron. Eng.* **2021**, *17*, 143–150. [\[CrossRef\]](#)
73. Dirik, M.; Kocamaz, F. Rrt-dijkstra: An improved path planning algorithm for mobile robots. *J. Soft Comput. Artif. Intell.* **2020**, *1*, 69–77.
74. Li, Y.; Scanavino, M.; Capello, E.; Dabbene, F.; Guglieri, G.; Vilardi, A. A novel distributed architecture for UAV indoor navigation. *Transp. Res. Procedia* **2018**, *35*, 13–22. [\[CrossRef\]](#)
75. Malang, C.; Charoenkwan, P.; Wudhikarn, R. Implementation and critical factors of unmanned aerial vehicle (UAV) in warehouse management: A systematic literature review. *Drones* **2023**, *7*, 80. [\[CrossRef\]](#)
76. Lu, Y.; Xue, Z.; Xia, G.S.; Zhang, L. A survey on vision-based UAV navigation. *Geo-Spat. Inf. Sci.* **2018**, *21*, 21–32. [\[CrossRef\]](#)
77. Aloqaily, M.; Hussain, R.; Khalaf, D.; Slehat, D.; Oracevic, A. On the role of futuristic technologies in securing UAV-supported autonomous vehicles. *IEEE Consum. Electron. Mag.* **2022**, *11*, 93–105. [\[CrossRef\]](#)
78. Faıçal, B.S.; Freitas, H.; Gomes, P.H.; Mano, L.Y.; Pessin, G.; de Carvalho, A.C.; Krishnamachari, B.; Ueyama, J. An adaptive approach for UAV-based pesticide spraying in dynamic environments. *Comput. Electron. Agric.* **2017**, *138*, 210–223. [\[CrossRef\]](#)
79. Chai, R.; Savvaris, A.; Tsourdos, A.; Chai, S.; Xia, Y. A review of optimization techniques in spacecraft flight trajectory design. *Prog. Aerosp. Sci.* **2019**, *109*, 100543. [\[CrossRef\]](#)
80. Liu, H.; Chen, Q.; Pan, N.; Sun, Y.; Yang, Y. Three-dimensional mountain complex terrain and heterogeneous multi-UAV cooperative combat mission planning. *IEEE Access* **2020**, *8*, 197407–197419. [\[CrossRef\]](#)
81. Ruzgienė, B.; Berteška, T.; Gečyte, S.; Jakubauskienė, E.; Aksamitauskas, V.Č. The surface modelling based on UAV Photogrammetry and qualitative estimation. *Measurement* **2015**, *73*, 619–627. [\[CrossRef\]](#)
82. Jung, S.; Kim, H. Analysis of amazon prime air uav delivery service. *J. Knowl. Inf. Technol. Syst.* **2017**, *12*, 253–266.
83. Nex, F.; Remondino, F. UAV for 3D mapping applications: A review. *Appl. Geomat.* **2014**, *6*, 1–15. [\[CrossRef\]](#)
84. Albani, D.; Nardi, D.; Trianni, V. Field coverage and weed mapping by UAV swarms. In Proceedings of the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Vancouver, BC, Canada, 24–28 September 2017; pp. 4319–4325.
85. Xu, C.; Xu, M.; Yin, C. Optimized multi-UAV cooperative path planning under the complex confrontation environment. *Comput. Commun.* **2020**, *162*, 196–203. [\[CrossRef\]](#)
86. Scherer, J.; Yahyanejad, S.; Hayat, S.; Yanmaz, E.; Andre, T.; Khan, A.; Vukadinovic, V.; Bettstetter, C.; Hellwagner, H.; Rinner, B. An autonomous multi-UAV system for search and rescue. In Proceedings of the First Workshop on Micro Aerial Vehicle Networks, Systems, and Applications for Civilian Use, Florence, Italy, 18 May 2015; pp. 33–38.
87. Ya’acob, N.; Zolkapli, M.; Johari, J.; Yusof, A.L.; Sarnin, S.S.; Asmadinar, A.Z. UAV environment monitoring system. In Proceedings of the 2017 International Conference on Electrical, Electronics and System Engineering (ICEESE), Kanazawa, Japan, 9–10 November 2017; pp. 105–109.
88. Noto, M.; Sato, H. A method for the shortest path search by extended Dijkstra algorithm. In Proceedings of the SMC 2000 Conference Proceedings—2000 IEEE International Conference on Systems, Man and Cybernetics, ‘Cybernetics Evolving to Systems, Humans, Organizations, and Their Complex Interactions’ (cat. no. 0), Nashville, TN, USA, 8–11 October 2000; Volume 3, pp. 2316–2320.
89. Zhou, Y.; Huang, N. Airport AGV path optimization model based on ant colony algorithm to optimize Dijkstra algorithm in urban systems. *Sustain. Comput. Inform. Syst.* **2022**, *35*, 100716. [\[CrossRef\]](#)
90. Xu, Y.; Wen, Z.; Zhang, X. Indoor optimal path planning based on Dijkstra Algorithm. In Proceedings of the International Conference on Materials Engineering and Information Technology Applications (MEITA 2015), Guangzhou, China, 17–18 January 2015; Atlantis Press: Dordrecht, The Netherlands, 2015; pp. 309–313.
91. Nash, A.; Daniel, K.; Koenig, S.; Felner, A. Theta*: Any-angle path planning on grids. In Proceedings of the Association for the Advancement of Artificial Intelligence (AAAI), Vancouver, BC, Canada, 22–26 July 2007; Volume 7, pp. 1177–1183.
92. Ferguson, D.; Stentz, A. Using interpolation to improve path planning: The Field D* algorithm. *J. Field Robot.* **2006**, *23*, 79–101. [\[CrossRef\]](#)
93. LaValle, S. Rapidly-exploring random trees: A new tool for path planning. *Annu. Res. Rep. TR 98-11* **1998**, *7*, 1177–1183.
94. Kavraki, L.E.; LaValle, S.M. Motion planning. In *Springer Handbook of Robotics*; Springer: Berlin/Heidelberg, Germany, 2016; pp. 139–162.
95. Kavraki, L.E.; Svestka, P.; Latombe, J.C.; Overmars, M.H. Probabilistic roadmaps for path planning in high-dimensional configuration spaces. *IEEE Trans. Robot. Autom.* **1996**, *12*, 566–580. [\[CrossRef\]](#)
96. LaValle, S.M. *Planning Algorithms*; Cambridge University Press: Cambridge, UK, 2006.
97. Bohlin, R.; Kavraki, L.E. Path planning using lazy PRM. In Proceedings of the 2000 ICRA, Millennium Conference, IEEE International Conference on Robotics and Automation, Symposia Proceedings (Cat. No. 00CH37065), San Francisco, CA, USA, 24–28 April 2000; Volume 1, pp. 521–528.
98. Naderi, K.; Rajamäki, J.; Hämäläinen, P. RT-RRT* a real-time path planning algorithm based on RRT. In Proceedings of the 8th ACM SIGGRAPH Conference on Motion in Games, Paris, France, 16–18 November 2015; pp. 113–118.

99. Gammell, J.D.; Srinivasa, S.S.; Barfoot, T.D. Informed RRT*: Optimal sampling-based path planning focused via direct sampling of an admissible ellipsoidal heuristic. In Proceedings of the 2014 IEEE/RSJ International Conference on Intelligent Robots and Systems, Chicago, IL, USA, 14–18 September 2014; pp. 2997–3004.
100. Islam, F.; Nasir, J.; Malik, U.; Ayaz, Y.; Hasan, O. Rrt*-smart: Rapid convergence implementation of rrt* towards optimal solution. In Proceedings of the 2012 IEEE International Conference on Mechatronics and Automation, Chengdu, China, 5–8 August 2012; pp. 1651–1656.
101. Wang, B.; Ju, D.; Xu, F.; Feng, C. Bi-RRT*: An Improved Bidirectional RRT* Path Planner for Robot in Two-Dimensional Space. *IEEJ Trans. Electr. Electron. Eng.* **2023**, *18*, 1639–1652. [\[CrossRef\]](#)
102. Kalisiak, M.; van de Panne, M. RRT-blossom: RRT with a local flood-fill behavior. In Proceedings of the 2006 IEEE International Conference on Robotics and Automation (ICRA), Orlando, FL, USA, 15–19 May 2006; pp. 1237–1242.
103. Meng, W.; Gong, Y.; Xu, F.; Tao, P.; Bo, P.; Xin, S. Efficient path planning for AUVs in unmapped marine environments using a hybrid local–global strategy. *Ocean Eng.* **2023**, *288*, 116227. [\[CrossRef\]](#)
104. Lian, J.; Cui, C.; Sun, W.; Wu, Y.; Huang, R. KD-RRT: Restricted Random Testing based on K-Dimensional Tree. In Proceedings of the 2021 8th International Conference on Dependable Systems and Their Applications (DSA), Yinchuan, China, 5–6 August 2021; pp. 590–599.
105. Zhao, P.; Chang, Y.; Wu, W.; Luo, H.; Zhou, Z.; Qiao, Y.; Li, Y.; Zhao, C.; Huang, Z.; Liu, B.; et al. Dynamic RRT: Fast feasible path planning in randomly distributed obstacle environments. *J. Intell. Robot. Syst.* **2023**, *107*, 48. [\[CrossRef\]](#)
106. Mashayekhi, R.; Idris, M.Y.I.; Anisi, M.H.; Ahmedy, I. Hybrid RRT: A semi-dual-tree RRT-based motion planner. *IEEE Access* **2020**, *8*, 18658–18668. [\[CrossRef\]](#)
107. Holland, J.H. Genetic algorithms. *Sci. Am.* **1992**, *267*, 66–73. [\[CrossRef\]](#)
108. Ramirez-Atencia, C.; Bello-Organ, G.; R-Moreno, M.D.; Camacho, D. Solving complex multi-UAV mission planning problems using multi-objective genetic algorithms. *Soft Comput.* **2017**, *21*, 4883–4900. [\[CrossRef\]](#)
109. Pradeepmon, T.; Sridharan, R.; Panicker, V.V. Development of modified discrete particle swarm optimization algorithm for quadratic assignment problems. *Int. J. Ind. Eng. Comput.* **2018**, *9*, 491–508. [\[CrossRef\]](#)
110. Jain, M.; Saihjal, V.; Singh, N.; Singh, S.B. An overview of variants and advancements of PSO algorithm. *Appl. Sci.* **2022**, *12*, 8392. [\[CrossRef\]](#)
111. Dorigo, M.; Birattari, M.; Stutzle, T. Ant colony optimization. *IEEE Comput. Intell. Mag.* **2007**, *1*, 28–39. [\[CrossRef\]](#)
112. Jiang, B.; Huang, G.; Wang, T.; Gui, J.; Zhu, X. Trust based energy efficient data collection with unmanned aerial vehicle in edge network. *Trans. Emerg. Telecommun. Technol.* **2022**, *33*, e3942. [\[CrossRef\]](#)
113. Dorigo, M.; Blum, C. Ant colony optimization theory: A survey. *Theor. Comput. Sci.* **2005**, *344*, 243–278. [\[CrossRef\]](#)
114. Minh, D.T.; Dung, N.B. Hybrid algorithms in path planning for autonomous navigation of unmanned aerial vehicle: A comprehensive review. *Meas. Sci. Technol.* **2024**, *35*, 112002. [\[CrossRef\]](#)
115. Ting, T.; Yang, X.S.; Cheng, S.; Huang, K. Hybrid metaheuristic algorithms: Past, present, and future. In *Recent Advances in Swarm Intelligence and Evolutionary Computation*; Springer: Cham, Switzerland, 2015; pp. 71–83.
116. Abdel-Basset, M.; Abdel-Fatah, L.; Sangaiah, A.K. Metaheuristic algorithms: A comprehensive review. In *Computational Intelligence for Multimedia Big Data on the Cloud with Engineering Applications*; Academic Press: Cambridge, MA, USA, 2018; pp. 185–231.
117. Chen, C.; Cao, L.; Chen, Y.; Chen, B.; Yue, Y. A comprehensive survey of convergence analysis of beetle antennae search algorithm and its applications. *Artif. Intell. Rev.* **2024**, *57*, 141. [\[CrossRef\]](#)
118. Wu, Q.; Shen, X.; Jin, Y.; Chen, Z.; Li, S.; Khan, A.H.; Chen, D. Intelligent beetle antennae search for UAV sensing and avoidance of obstacles. *Sensors* **2019**, *19*, 1758. [\[CrossRef\]](#)
119. Kang, Y.; Hedrick, J.K. Linear tracking for a fixed-wing UAV using nonlinear model predictive control. *IEEE Trans. Control Syst. Technol.* **2009**, *17*, 1202–1210. [\[CrossRef\]](#)
120. Mkiramweni, M.E.; Yang, C.; Li, J.; Zhang, W. A survey of game theory in unmanned aerial vehicles communications. *IEEE Commun. Surv. Tutor.* **2019**, *21*, 3386–3416. [\[CrossRef\]](#)
121. Cui, Z.; Wang, Y. UAV path planning based on multi-layer reinforcement learning technique. *IEEE Access* **2021**, *9*, 59486–59497. [\[CrossRef\]](#)
122. Afifi, G.; Gadallah, Y. Cellular Network-Supported Machine Learning Techniques for Autonomous UAV Trajectory Planning. *IEEE Access* **2022**, *10*, 131996–132011. [\[CrossRef\]](#)
123. Qu, C.; Gai, W.; Zhong, M.; Zhang, J. A novel reinforcement learning based grey wolf optimizer algorithm for unmanned aerial vehicles (UAVs) path planning. *Appl. Soft Comput.* **2020**, *89*, 106099. [\[CrossRef\]](#)
124. Wang, Z.; Ng, S.X.; El-Hajjar, M. A 3D Spatial Information Compression Based Deep Reinforcement Learning Technique for UAV Path Planning in Cluttered Environments. *IEEE Open J. Veh. Technol.* **2025**, *6*, 647–661. [\[CrossRef\]](#)
125. Sanyal, S.; Joshi, A.; Nagaraj, M.; Manna, R.K.; Roy, K. Energy-Efficient Autonomous Aerial Navigation with Dynamic Vision Sensors: A Physics-Guided Neuromorphic Approach. *arXiv* **2025**, arXiv:2502.05938.

126. Mahesh, B. Machine learning algorithms—A review. *Int. J. Sci. Res. (IJSR)* **2020**, *9*, 381–386. [\[CrossRef\]](#)
127. Cunningham, P.; Cord, M.; Delany, S.J. Supervised learning. In *Machine Learning Techniques for Multimedia: Case Studies on Organization and Retrieval*; Springer: Berlin/Heidelberg, Germany, 2008; pp. 21–49.
128. Hastie, T.; Tibshirani, R.; Friedman, J.; Hastie, T.; Tibshirani, R.; Friedman, J. Overview of supervised learning. In *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*; Springer: New York, NY, USA, 2009; pp. 9–41.
129. Miura, J. Support vector path planning. In Proceedings of the 2006 IEEE/RSJ International Conference on Intelligent Robots and Systems, Beijing, China, 9–15 October 2006; pp. 2894–2899.
130. Chen, Y.; Zu, W.; Fan, G.; Chang, H. Unmanned aircraft vehicle path planning based on SVM algorithm. In *Foundations and Practical Applications of Cognitive Systems and Information Processing*; Springer: Berlin/Heidelberg, Germany, 2014; pp. 705–714.
131. Moon, B.; Akagi, C.; Peterson, C.K. Decentralized Multi-Agent Search for Moving Targets Using Road Network Gaussian Process Regressions. *Drones* **2024**, *8*, 606. [\[CrossRef\]](#)
132. Zhang, C.; Wang, Y.; Zheng, W. Multi-UAVs Tracking Non-Cooperative Target Using Constrained Iterative Linear Quadratic Gaussian. *Drones* **2024**, *8*, 326. [\[CrossRef\]](#)
133. Almazrouei, K.; Kamel, I.; Rabie, T. Dynamic obstacle avoidance and path planning through reinforcement learning. *Appl. Sci.* **2023**, *13*, 8174. [\[CrossRef\]](#)
134. Hong, D.; Lee, S.; Cho, Y.H.; Baek, D.; Kim, J.; Chang, N. Energy-efficient online path planning of multiple drones using reinforcement learning. *IEEE Trans. Veh. Technol.* **2021**, *70*, 9725–9740. [\[CrossRef\]](#)
135. Li, B.; Wu, Y. Path planning for UAV ground target tracking via deep reinforcement learning. *IEEE Access* **2020**, *8*, 29064–29074. [\[CrossRef\]](#)
136. He, L.; Aouf, N.; Song, B. Explainable Deep Reinforcement Learning for UAV autonomous path planning. *Aerosp. Sci. Technol.* **2021**, *118*, 107052. [\[CrossRef\]](#)
137. Zhang, D.; Xuan, Z.; Zhang, Y.; Yao, J.; Li, X.; Li, X. Path planning of unmanned aerial vehicle in complex environments based on state-detection twin delayed deep deterministic policy gradient. *Machines* **2023**, *11*, 108. [\[CrossRef\]](#)
138. Wu, X.; Huang, S.; Huang, G. Deep reinforcement learning-based 2.5 D multi-objective path planning for ground vehicles: Considering distance and energy consumption. *Electronics* **2023**, *12*, 3840. [\[CrossRef\]](#)
139. Kaelbling, L.P.; Littman, M.L.; Moore, A.W. Reinforcement learning: A survey. *J. Artif. Intell. Res.* **1996**, *4*, 237–285. [\[CrossRef\]](#)
140. Wiering, M.A.; Van Otterlo, M. Reinforcement learning. *Adapt. Learn. Optim.* **2012**, *12*, 729.
141. Yan, C.; Xiang, X. A path planning algorithm for UAV based on improved Q-learning. In Proceedings of the 2018 2nd International Conference on Robotics and Automation Sciences (ICRAS), Wuhan, China, 23–25 June 2018; pp. 1–5.
142. Yang, Y.; Juntao, L.; Lingling, P. Multi-robot path planning based on a deep reinforcement learning DQN algorithm. *CAAI Trans. Intell. Technol.* **2020**, *5*, 177–183. [\[CrossRef\]](#)
143. Yan, C.; Xiang, X.; Wang, C. Towards real-time path planning through deep reinforcement learning for a UAV in dynamic environments. *J. Intell. Robot. Syst.* **2020**, *98*, 297–309. [\[CrossRef\]](#)
144. Tian, S.; Li, Y.; Zhang, X.; Zheng, L.; Cheng, L.; She, W.; Xie, W. Fast UAV path planning in urban environments based on three-step experience buffer sampling DDPG. *Digit. Commun. Netw.* **2024**, *10*, 813–826. [\[CrossRef\]](#)
145. Bouhamed, O.; Ghazzai, H.; Besbes, H.; Massoud, Y. Autonomous UAV navigation: A DDPG-based deep reinforcement learning approach. In Proceedings of the 2020 IEEE International Symposium on Circuits and Systems (ISCAS), Seville, Spain, 12–14 October 2020; pp. 1–5.
146. LeCun, Y.; Bengio, Y.; Hinton, G. Deep learning. *Nature* **2015**, *521*, 436–444. [\[CrossRef\]](#) [\[PubMed\]](#)
147. Pan, Y.; Yang, Y.; Li, W. A deep learning trained by genetic algorithm to improve the efficiency of path planning for data collection with multi-UAV. *IEEE Access* **2021**, *9*, 7994–8005. [\[CrossRef\]](#)
148. Bayerlein, H.; Theile, M.; Caccamo, M.; Gesbert, D. Multi-UAV path planning for wireless data harvesting with deep reinforcement learning. *IEEE Open J. Commun. Soc.* **2021**, *2*, 1171–1187. [\[CrossRef\]](#)
149. Castro, G.G.d.; Berger, G.S.; Cantieri, A.; Teixeira, M.; Lima, J.; Pereira, A.I.; Pinto, M.F. Adaptive path planning for fusing rapidly exploring random trees and deep reinforcement learning in an agriculture dynamic environment UAVs. *Agriculture* **2023**, *13*, 354. [\[CrossRef\]](#)
150. Voulodimos, A.; Doulamis, N.; Doulamis, A.; Protopapadakis, E. Deep learning for computer vision: A brief review. *Comput. Intell. Neurosci.* **2018**, *2018*, 7068349. [\[CrossRef\]](#)
151. Li, Y.; Ma, L.; Zhong, Z.; Liu, F.; Chapman, M.A.; Cao, D.; Li, J. Deep learning for lidar point clouds in autonomous driving: A review. *IEEE Trans. Neural Netw. Learn. Syst.* **2020**, *32*, 3412–3432. [\[CrossRef\]](#)
152. Zhang, L.; Zhang, L.; Du, B. Deep learning for remote sensing data: A technical tutorial on the state of the art. *IEEE Geosci. Remote Sens. Mag.* **2016**, *4*, 22–40. [\[CrossRef\]](#)
153. Sze, V.; Chen, Y.H.; Yang, T.J.; Emer, J.S. Efficient processing of deep neural networks: A tutorial and survey. *Proc. IEEE* **2017**, *105*, 2295–2329. [\[CrossRef\]](#)

154. Zhang, Z.; Wang, S.; Chen, J.; Han, Y. A bionic dynamic path planning algorithm of the micro UAV based on the fusion of deep neural network optimization/filtering and hawk-eye vision. *IEEE Trans. Syst. Man Cybern. Syst.* **2023**, *53*, 3728–3740. [\[CrossRef\]](#)
155. Akshya, J.; Neelamegam, G.; Sureshkumar, C.; Nithya, V.; Kadry, S. Enhancing UAV Path Planning Efficiency through Adam-Optimized Deep Neural Networks for Area Coverage Missions. *Procedia Comput. Sci.* **2024**, *235*, 2–11.
156. Sivaranjani, A.; Vinod, B. Artificial Potential Field Incorporated Deep-Q-Network Algorithm for Mobile Robot Path Prediction. *Intell. Autom. Soft Comput.* **2023**, *35*, 1745–1760. [\[CrossRef\]](#)
157. Cuomo, S.; Di Cola, V.S.; Giampaolo, F.; Rozza, G.; Raissi, M.; Piccialli, F. Scientific machine learning through physics-informed neural networks: Where we are and what's next. *J. Sci. Comput.* **2022**, *92*, 88. [\[CrossRef\]](#)
158. Kyrkou, C.; Plastiras, G.; Theodoridis, T.; Venieris, S.I.; Bouganis, C.S. DroNet: Efficient convolutional neural network detector for real-time UAV applications. In Proceedings of the 2018 Design, Automation & Test in Europe Conference & Exhibition (DATE), Dresden, Germany, 19–23 March 2018; pp. 967–972.
159. O'Shea, K. An introduction to convolutional neural networks. *arXiv* **2015**, arXiv:1511.08458.
160. Bae, H.; Kim, G.; Kim, J.; Qian, D.; Lee, S. Multi-robot path planning method using reinforcement learning. *Appl. Sci.* **2019**, *9*, 3057. [\[CrossRef\]](#)
161. Li, S.; Qian, Y. Unmanned Aerial Vehicle Autonomous Flight Path Planning Algorithm Based on Deep Learning. In Proceedings of the 2024 International Conference on Telecommunications and Power Electronics (TELEPE), Frankfurt, Germany, 29–31 May 2024; pp. 632–637.
162. Porzi, L.; Rota Bulò, S.; Lepri, B.; Ricci, E. Predicting and understanding urban perception with convolutional neural networks. In Proceedings of the 23rd ACM International Conference on Multimedia, Brisbane, Australia, 26–30 October 2015; pp. 139–148.
163. McGuire, K.N.; De Wagter, C.; Tuyls, K.; Kappen, H.J.; de Croon, G.C. Minimal navigation solution for a swarm of tiny flying robots to explore an unknown environment. *Sci. Robot.* **2019**, *4*, eaaw9710. [\[CrossRef\]](#)
164. Creswell, A.; White, T.; Dumoulin, V.; Arulkumaran, K.; Sengupta, B.; Bharath, A.A. Generative adversarial networks: An overview. *IEEE Signal Process. Mag.* **2018**, *35*, 53–65. [\[CrossRef\]](#)
165. Ma, N.; Wang, J.; Liu, J.; Meng, M.Q.H. Conditional generative adversarial networks for optimal path planning. *IEEE Trans. Cogn. Dev. Syst.* **2021**, *14*, 662–671. [\[CrossRef\]](#)
166. Eskandari, M.; Savkin, A.V. GANs the UAV Path Planner: UAV-Based RIS-Assisted Wireless Communication for Internet of Autonomous Vehicles. In Proceedings of the 2024 IEEE 19th Conference on Industrial Electronics and Applications (ICIEA), Kristiansand, Norway, 5–8 August 2024; pp. 1–6.
167. Gan, J.; Li, M.; Li, Q.; Zhang, R.; Zheng, X. IUAV Path Planning Using a Multiobjective Projection Algorithm. *IEEE Trans. Ind. Inform.* **2024**, *20*, 13069–13076. [\[CrossRef\]](#)
168. Mohammadi, M.; Al-Fuqaha, A.; Oh, J.S. Path planning in support of smart mobility applications using generative adversarial networks. In Proceedings of the 2018 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData), Halifax, NS, Canada, 30 July–3 August 2018; pp. 878–885.
169. Yan, H.; Chen, Y.; Yang, S.H. New energy consumption model for rotary-wing UAV propulsion. *IEEE Wirel. Commun. Lett.* **2021**, *10*, 2009–2012. [\[CrossRef\]](#)
170. Na, Y.; Li, Y.; Chen, D.; Yao, Y.; Li, T.; Liu, H.; Wang, K. Optimal energy consumption path planning for unmanned aerial vehicles based on improved particle swarm optimization. *Sustainability* **2023**, *15*, 12101. [\[CrossRef\]](#)
171. Cabreira, T.M.; Di Franco, C.; Ferreira, P.R.; Buttazzo, G.C. Energy-aware spiral coverage path planning for uav photogrammetric applications. *IEEE Robot. Autom. Lett.* **2018**, *3*, 3662–3668. [\[CrossRef\]](#)
172. Palossi, D.; Furci, M.; Naldi, R.; Marongiu, A.; Marconi, L.; Benini, L. An energy-efficient parallel algorithm for real-time near-optimal uav path planning. In Proceedings of the ACM International Conference on Computing Frontiers, Como, Italy, 16–19 May 2016; pp. 392–397.
173. Yacef, F.; Rizoug, N.; Degaa, L.; Hamerlain, M. Energy-efficiency path planning for quadrotor UAV under wind conditions. In Proceedings of the 2020 7th International Conference on Control, Decision and Information Technologies (CoDIT), Prague, Czech Republic, 29 June–2 July 2020; Volume 1, pp. 1133–1138.
174. Sariççek, İ.; Akkuş, Y. Unmanned aerial vehicle hub-location and routing for monitoring geographic borders. *Appl. Math. Model.* **2015**, *39*, 3939–3953. [\[CrossRef\]](#)
175. Yu, V.F.; Lin, S.Y. Solving the location-routing problem with simultaneous pickup and delivery by simulated annealing. *Int. J. Prod. Res.* **2016**, *54*, 526–549. [\[CrossRef\]](#)
176. Tian, P.; Chao, H.; Rhudy, M.; Gross, J.; Wu, H. Wind sensing and estimation using small fixed-wing unmanned aerial vehicles: A survey. *J. Aerosp. Inf. Syst.* **2021**, *18*, 132–143. [\[CrossRef\]](#)
177. Shamshad, A.; Bawadi, M.; Hussin, W.W.; Majid, T.A.; Sanusi, S. First and second order Markov chain models for synthetic generation of wind speed time series. *Energy* **2005**, *30*, 693–708. [\[CrossRef\]](#)

178. Negra, N.B.; Holmstrøm, O.; Bak-Jensen, B.; Sørensen, P. Model of a synthetic wind speed time series generator. *Wind Energy Int. J. Prog. Appl. Wind Power Convers. Technol.* **2008**, *11*, 193–209. [\[CrossRef\]](#)
179. Li, Y.; Liu, M. Path planning of electric VTOL UAV considering minimum energy consumption in urban areas. *Sustainability* **2022**, *14*, 13421. [\[CrossRef\]](#)
180. Dorling, K.; Heinrichs, J.; Messier, G.G.; Magierowski, S. Vehicle routing problems for drone delivery. *IEEE Trans. Syst. Man Cybern. Syst.* **2016**, *47*, 70–85. [\[CrossRef\]](#)
181. Kim, S.J.; Lim, G.J.; Cho, J. Drone flight scheduling under uncertainty on battery duration and air temperature. *Comput. Ind. Eng.* **2018**, *117*, 291–302. [\[CrossRef\]](#)
182. Yan, X.; Chen, R.; Jiang, Z. UAV Cluster Mission Planning Strategy for Area Coverage Tasks. *Sensors* **2023**, *23*, 9122. [\[CrossRef\]](#) [\[PubMed\]](#)
183. Makam, S.; Komatineni, B.K.; Meena, S.S.; Meena, U. Unmanned aerial vehicles (UAVs): An adoptable technology for precise and smart farming. *Discov. Internet Things* **2024**, *4*, 12. [\[CrossRef\]](#)
184. Ghauri, S.A.; Sarfraz, M.; Qamar, R.A.; Sohail, M.F.; Khan, S.A. A Review of Multi-UAV Task Allocation Algorithms for a Search and Rescue Scenario. *J. Sens. Actuator Netw.* **2024**, *13*, 47. [\[CrossRef\]](#)
185. Wang, W.; Zhang, G.; Da, Q.; Lu, D.; Zhao, Y.; Li, S.; Lang, D. Multiple unmanned aerial vehicle autonomous path planning algorithm based on whale-inspired deep Q-network. *Drones* **2023**, *7*, 572. [\[CrossRef\]](#)
186. Choi, H.L.; Brunet, L.; How, J.P. Consensus-based decentralized auctions for robust task allocation. *IEEE Trans. Robot.* **2009**, *25*, 912–926. [\[CrossRef\]](#)
187. Qamar, R.A.; Sarfraz, M.; Ghauri, S.A.; Baig, N.A.; Cheema, T.A. Optimization of Dynamic Task Allocation for Multi-UAV Systems: Search and Rescue Scenario. 2024. Available online: <https://www.researchsquare.com/article/rs-3879027/v1> (accessed on 24 January 2024).
188. Meng, K.; He, X.; Wu, Q.; Li, D. Multi-UAV collaborative sensing and communication: Joint task allocation and power optimization. *IEEE Trans. Wirel. Commun.* **2022**, *22*, 4232–4246. [\[CrossRef\]](#)
189. Zhang, P.; He, Y.; Wang, Z.; Li, S.; Liang, Q. Research on Multi-UAV Obstacle Avoidance with Optimal Consensus Control and Improved APF. *Drones* **2024**, *8*, 248. [\[CrossRef\]](#)
190. Luo, W.; Tang, Q.; Fu, C.; Eberhard, P. Deep-sarsa based multi-UAV path planning and obstacle avoidance in a dynamic environment. In Proceedings of the Advances in Swarm Intelligence: 9th International Conference, ICSI 2018, Shanghai, China, 17–22 June 2018; Proceedings, Part II 9; Springer: Berlin/Heidelberg, Germany, 2018; pp. 102–111.
191. Wang, L.; Huang, W.; Li, H.; Li, W.; Chen, J.; Wu, W. A review of collaborative trajectory planning for multiple unmanned aerial vehicles. *Processes* **2024**, *12*, 1272. [\[CrossRef\]](#)
192. Khan, A.A.; Khan, M.M.; Khan, K.M.; Arshad, J.; Ahmad, F. A blockchain-based decentralized machine learning framework for collaborative intrusion detection within UAVs. *Comput. Netw.* **2021**, *196*, 108217. [\[CrossRef\]](#)
193. Mao, A.; Tang, X.; Ding, Z.; Li, Z. Scalability optimization of centralized cluster resource scheduling framework. *J. Comput. Res. Dev.* **2021**, *58*, 497–512.
194. Liu, J.; Zhang, Z.; Xu, T.; Yan, C. Multi-UAV United Task Allocation via Extended Market Mechanism Based on Flight Path Cost. In Proceedings of the 2024 IEEE International Conference on Unmanned Systems (ICUS), Nanjing, China, 18–20 October 2024; pp. 50–55.
195. Zhu, A.; Yang, S.X. A neural network approach to dynamic task assignment of multirobots. *IEEE Trans. Neural Netw.* **2006**, *17*, 1278–1287. [\[PubMed\]](#)
196. Dai, W.; Lu, H.; Xiao, J.; Zeng, Z.; Zheng, Z. Multi-robot dynamic task allocation for exploration and destruction. *J. Intell. Robot. Syst.* **2020**, *98*, 455–479. [\[CrossRef\]](#)
197. Xiong, T.; Liu, F.; Liu, H.; Ge, J.; Li, H.; Ding, K.; Li, Q. Multi-drone optimal mission assignment and 3D path planning for disaster rescue. *Drones* **2023**, *7*, 394. [\[CrossRef\]](#)
198. Chandran, I.; Vipin, K. Multi-UAV networks for disaster monitoring: Challenges and opportunities from a network perspective. *Drone Syst. Appl.* **2024**, *12*, 1–28.
199. Muñoz, J.; López, B.; Quevedo, F.; Monje, C.A.; Garrido, S.; Moreno, L.E. Multi UAV coverage path planning in urban environments. *Sensors* **2021**, *21*, 7365. [\[CrossRef\]](#)
200. Wu, Q.; Su, Y.; Tan, W.; Zhan, R.; Liu, J.; Jiang, L. UAV Path Planning Trends from 2000 to 2024: A Bibliometric Analysis and Visualization. *Drones* **2025**, *9*, 128. [\[CrossRef\]](#)
201. Liu, J.; Yan, Y.; Yang, Y.; Li, J. An improved artificial potential field UAV path planning algorithm guided by RRT under environment-aware modeling: Theory and simulation. *IEEE Access* **2024**, *12*, 12080–12097. [\[CrossRef\]](#)
202. Zhang, W.; Zhang, S.; Wu, F.; Wang, Y. Path planning of UAV based on improved adaptive grey wolf optimization algorithm. *IEEE Access* **2021**, *9*, 89400–89411. [\[CrossRef\]](#)
203. Ramezani, M.; Amiri Atashgah, M.; Rezaee, A. A Fault-Tolerant Multi-Agent Reinforcement Learning Framework for Unmanned Aerial Vehicles–Unmanned Ground Vehicle Coverage Path Planning. *Drones* **2024**, *8*, 537. [\[CrossRef\]](#)

204. McEnroe, P.; Wang, S.; Liyanage, M. A survey on the convergence of edge computing and AI for UAVs: Opportunities and challenges. *IEEE Internet Things J.* **2022**, *9*, 15435–15459. [\[CrossRef\]](#)
205. Zhang, T.; Xu, Y.; Loo, J.; Yang, D.; Xiao, L. Joint computation and communication design for UAV-assisted mobile edge computing in IoT. *IEEE Trans. Ind. Inform.* **2019**, *16*, 5505–5516. [\[CrossRef\]](#)
206. Zhou, F.; Wu, Y.; Sun, H.; Chu, Z. UAV-enabled mobile edge computing: Offloading optimization and trajectory design. In Proceedings of the 2018 IEEE International Conference on Communications (ICC), Kansas City, MO, USA, 20–24 May 2018; pp. 1–6.
207. Xu, J.; Yao, H.; Zhang, R.; Mai, T.; Huang, S.; Guo, S. Federated learning powered semantic communication for UAV swarm cooperation. *IEEE Wirel. Commun.* **2024**, *31*, 140–146. [\[CrossRef\]](#)
208. Lim, W.Y.B.; Garg, S.; Xiong, Z.; Zhang, Y.; Niyato, D.; Leung, C.; Miao, C. UAV-assisted communication efficient federated learning in the era of the artificial intelligence of things. *IEEE Netw.* **2021**, *35*, 188–195. [\[CrossRef\]](#)
209. Balestrieri, E.; Daponte, P.; De Vito, L.; Picariello, F.; Tudosa, I. Sensors and measurements for UAV safety: An overview. *Sensors* **2021**, *21*, 8253. [\[CrossRef\]](#)
210. Khan, A.A.; Laghari, A.A.; Awan, S.A. Machine learning in computer vision: A review. *EAI Endorsed Trans. Scalable Inf. Syst.* **2021**, *8*, 1–11. [\[CrossRef\]](#)
211. Al-Fraihat, D.; Sharrab, Y.; Alzyoud, F.; Qahmash, A.; Tarawneh, M.; Maaita, A. Speech recognition utilizing deep learning: A systematic review of the latest developments. *Hum.-Centric Comput. Inf. Sci.* **2024**, *14*, 1–34. [\[CrossRef\]](#)
212. Luo, X.; Wang, Q.; Gong, H.; Tang, C. UAV path planning based on the average TD3 algorithm with prioritized experience replay. *IEEE Access* **2024**, *12*, 38017–38029. [\[CrossRef\]](#)
213. Abdalmanan, N.; Kamarudin, K.; Bakar, M.A.A.; Rahiman, M.H.F.; Zakaria, A.; Mamduh, S.M.; Kamarudin, L.M. 2D lidar based reinforcement learning for multi-target path planning in unknown environment. *IEEE Access* **2023**, *11*, 35541–35555. [\[CrossRef\]](#)
214. Harun, M.H.; Abdullah, S.S.; Aras, M.S.M.; Bahar, M.B. Sensor fusion technology for unmanned autonomous vehicles (UAV): A review of methods and applications. In Proceedings of the 2022 IEEE 9th International Conference on Underwater System Technology: Theory and Applications (USYS), Kuala Lumpur, Malaysia, 5–6 December 2022; pp. 1–8.
215. Derrouaoui, S.H.; Bouzid, Y.; Guiatni, M.; Dib, I. A comprehensive review on reconfigurable drones: Classification, characteristics, design and control technologies. *Unmanned Syst.* **2022**, *10*, 3–29. [\[CrossRef\]](#)
216. Gianfelice, M.; Aboshosha, H.; Ghazal, T. Real-time wind predictions for safe drone flights in Toronto. *Results Eng.* **2022**, *15*, 100534. [\[CrossRef\]](#)
217. Youn, W.; Ko, H.; Choi, H.; Choi, I.; Baek, J.H.; Myung, H. Collision-free autonomous navigation of a small UAV using low-cost sensors in GPS-denied environments. *Int. J. Control Autom. Syst.* **2021**, *19*, 953–968. [\[CrossRef\]](#)
218. Radwan, A.; Tourani, A.; Bavle, H.; Voos, H.; Sanchez-Lopez, J.L. UAV-assisted Visual SLAM Generating Reconstructed 3D Scene Graphs in GPS-denied Environments. In Proceedings of the 2024 International Conference on Unmanned Aircraft Systems (ICUAS), Chania, Crete, Greece, 4–7 June 2024; pp. 1109–1116.
219. Siwek, M.; Baranowski, L.; Ladyżyńska-Kozdraś, E. The Application and Optimisation of a Neural Network PID Controller for Trajectory Tracking Using UAVs. *Sensors* **2024**, *24*, 8072. [\[CrossRef\]](#) [\[PubMed\]](#)
220. Aljalaud, F.; Alohali, Y. Optimizing Autonomous Multi-UAV Path Planning for Inspection Missions: A Comparative Study of Genetic and Stochastic Hill Climbing Algorithms. *Energies* **2024**, *18*, 50. [\[CrossRef\]](#)
221. De Petrillo, M.; Beard, J.; Gu, Y.; Gross, J.N. Search planning of a uav/ugv team with localization uncertainty in a subterranean environment. *IEEE Aerosp. Electron. Syst. Mag.* **2021**, *36*, 6–16. [\[CrossRef\]](#)
222. Li, J.; Deng, G.; Luo, C.; Lin, Q.; Yan, Q.; Ming, Z. A hybrid path planning method in unmanned air/ground vehicle (UAV/UGV) cooperative systems. *IEEE Trans. Veh. Technol.* **2016**, *65*, 9585–9596. [\[CrossRef\]](#)
223. Nowakowski, M.; Berger, G.S.; Braun, J.; Mendes, J.a.; Bonzatto Junior, L.; Lima, J. Advance Reconnaissance of UGV Path Planning Using Unmanned Aerial Vehicle to Carry Our Mission in Unknown Environment. In *Robot 2023: Sixth Iberian Robotics Conference*; Springer: Cham, Switzerland, 2023; pp. 50–61.
224. Hu, D.; Gan, V.J.; Wang, T.; Ma, L. Multi-agent robotic system (MARS) for UAV-UGV path planning and automatic sensory data collection in cluttered environments. *Build. Environ.* **2022**, *221*, 109349. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.